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IGCSE Physics CIE

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1.7 Energy, Work & Power

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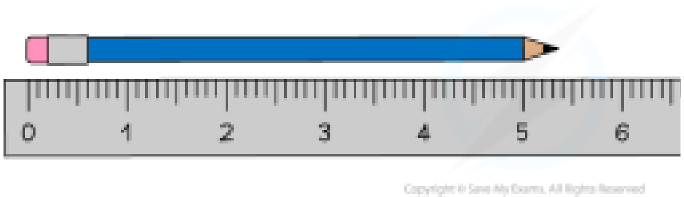
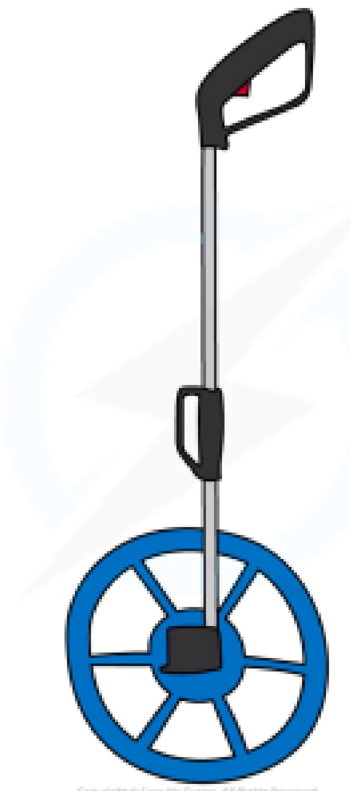
Measuring Length & Volume

Rulers can be used to measure small distances of a few centimetres (cm). They are able to measure to the nearest millimetre (mm)



A ruler can measure small distances to the nearest mm

When measuring larger distances (of a few metres) a **tape measure** is more appropriate or, when measuring even larger distances, a **trundle wheel**



Trundle wheels can be used to measure large distances

Measuring cylinders can be used to measure the volume of liquids

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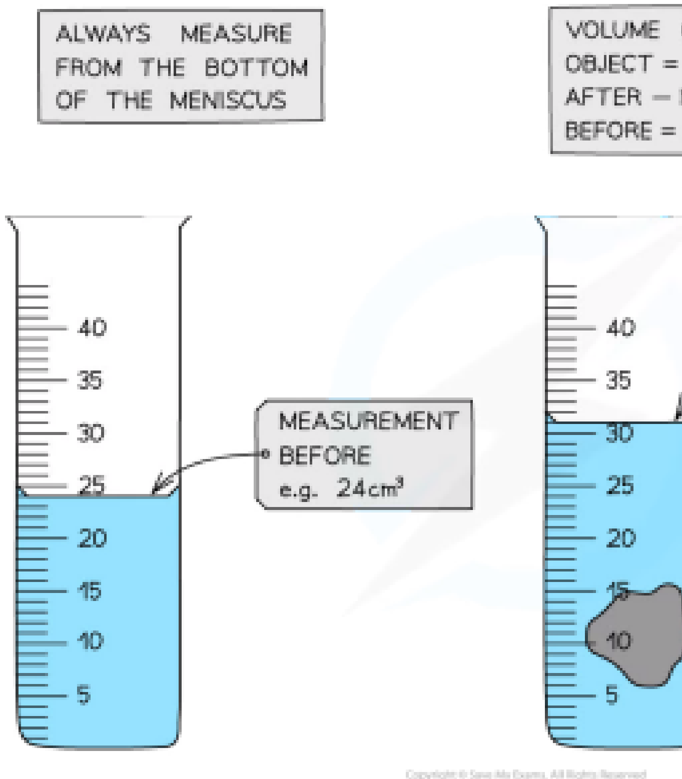
also be used to find the volume of an irregular shape

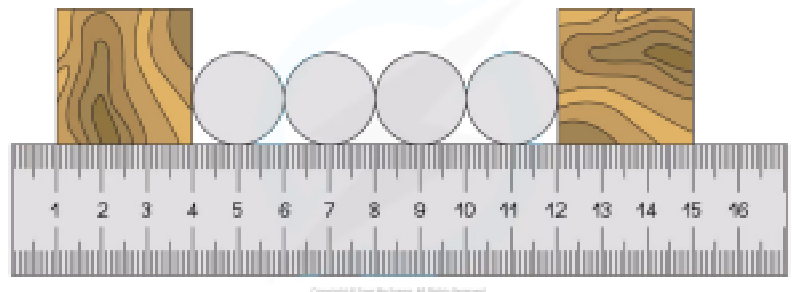
By measuring the **change in volume**, a measuring cylinder can

Measuring cylinders can be used to determine the volume of a liquid or an irregular shaped solid



Worked Example

The diagram shows four identical ball-bearings placed between two blocks on a steel ruler.



Calculate the diameter of one ball-bearing.

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Step 1: Measure the length of all four ball-bearings

The blocks mark the edges of the first and last ball bearings
The blocks make it easier to measure the length of all four

ball-bearings

Total length = 12 cm − 4 cm = **8 cm**

Step 2: Divide the total length by the number of ball-bearings Diameter = total length ÷ number of ball-bearings

Diameter = 8 ÷ 4

Diameter = **2 cm**

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Measuring Time

Stop-clocks and stopwatches can be used to measure time intervals An important factor when measuring time intervals is human reaction time. This can have a significant impact upon measurements when the measurements involved are very short (less than a second)

Worked Example

A stopwatch is used to measure the time taken for a runner to complete a lap of a 400 m track.

The images below give the readings on the stopwatch at the start and the end of the lap.

Calculate how long it took the runner to complete the lap. Give your answer in seconds.

Step 1: Identify the start time for the lap

The stopwatch was already at 0:55:10 when the runner started the lap Start time = **55.10 seconds (s)**

Step 2: Identify the finish time for the lap

The stopwatch reads 1:45:10 at the end of the lap
Finish time = **1 minute and 45.10 s**

Step 3: Convert the finish time into seconds

1 minute = 60 seconds

Finish time = 60 s + 45.10 s

Finish time = **105.10 s**

Step 4: Calculate the time taken to complete the lap

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The time taken to complete the lap = finish time − start time

Time taken to complete lap = 105.10 s − 55.10 s

Time taken to complete lap = **50 s**

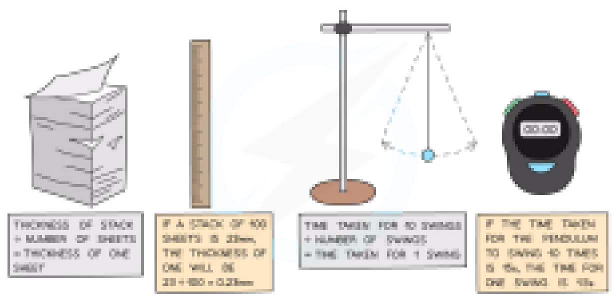
Exam Tip

You will sometimes find that information is given in the question that is not actually needed in the calculation.

In this worked example, you were told that the track the runner is running on is 400 m. This had nothing to do with the calculation the question asked you to perform.

This is a common method for making a question seem more difficult. Don't let it catch you out!

Multiple Readings



Suppose you have to measure the thickness of a sheet of paper The thing that you are
- trying to measure is so small that it would be very difficult to get an accurate answer

If, however, you measure the thickness of 100 sheets of paper you can do so much more accurately
- Dividing your answer by 100 will then give an accurate figure for the thickness of one sheet

This process of taking a reading of a large number of values and then dividing by the number, is a good way of getting accurate values for small figures, including (for example) the time period of a pendulum
- Measure the time taken for 10 swings and then **divide that time by 10** to find the **average**

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Scalar & Vector Quantities

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- All quantities can be one of two types:
  - A **scalar**
  - A **vector**

Scalars

- Scalars are quantities that have only a **magnitude**
  - For example, **mass** is a scalar since it is a quantity that has magnitude without a direction
  - **Distance** is also a scalar since it only contains a magnitude, not a direction

Vectors

- Vectors have both **magnitude** and **direction**
- **Velocity**, for instance, is a vector since it is described with both a magnitude and a direction
  - When describing the velocity of a car it is necessary to mention **both** its **speed** and the **direction** in which it is travelling
  - For example, the velocity might be 60 km per hour (magnitude) due west (direction)

- **Distance** is a value describing only how long an object is or how far it is between two points - this means it is a **scalar** quantity
- **Displacement** on the other hand also describes the **direction** in which the distance is measured - this means it is a **vector** quantity
  - For example, a displacement might be 100 km north

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Examples of Scalars & Vectors

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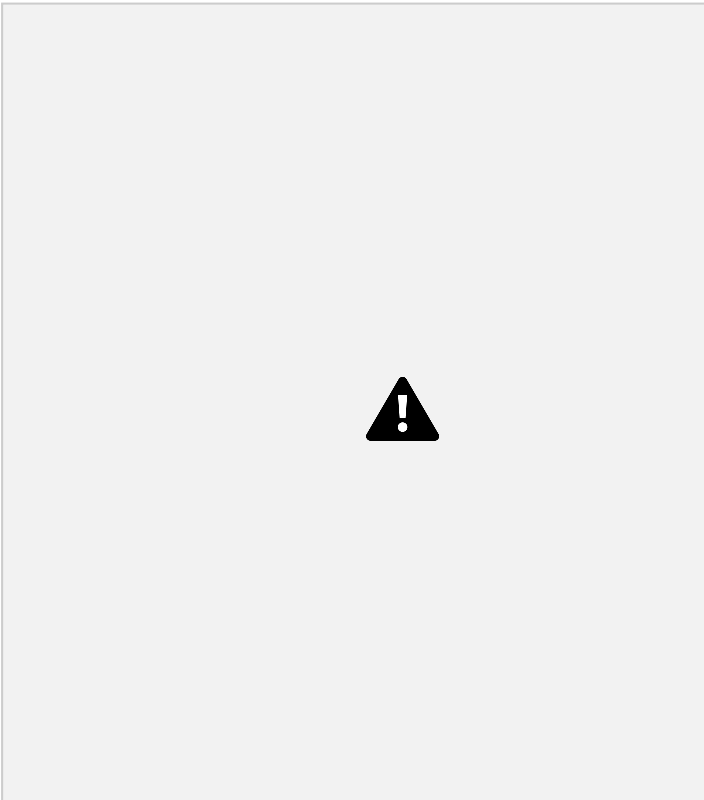
The table below lists some common examples of scalar and vector quantities: **Scalars & Vectors Table**



Some vectors and scalars are similar to each other

For example, the scalar quantity **distance** corresponds to the vector quantity **displacement**

Corresponding vectors and their scalar counterparts are aligned in the table where applicable

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Using Scalars & Vectors

The worked example below illustrates how to determine whether a quantity is a scalar or a vector

Worked Example

Blu is in charge of training junior astronauts. For one of his sessions, he would like to explain the difference between mass and weight.

Suggest how Blu should explain the difference between mass and weight, using definitions of scalars and vectors in your answer.

Step 1: Recall the definitions of a scalar and vector quantity

Scalars are quantities that have only a **magnitude**
Vectors are quantities that have both **magnitude and direction**

Step 2: Identify which quantity has magnitude only

Mass is a quantity with **magnitude** only
So mass is a **scalar** quantity
Blu might explain to his junior astronauts that their mass will not change if they travel to outer space

Step 3: Identify which quantity has magnitude and direction

Weight is a quantity with **magnitude and direction** (it is a force)
So weight is a **vector** quantity
Blu might explain that to his junior astronauts that their weight - the force on them due to gravity - will vary depending on their distance from the centre of the Earth

Exam Tip

Make sure you are comfortable with the differences between **similar** scalars and vectors, the most commonly confused pairings tend to be:

Distance and displacement
Speed and velocity
Weight and mass

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Calculations with Vectors

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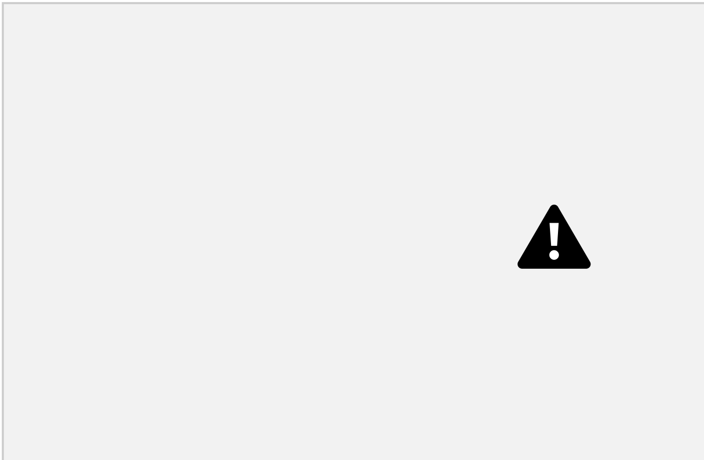
Vectors are represented by an arrow
The **arrowhead** indicates the **direction** of the vector
The **length** of the arrow represents the **magnitude**



The two force vectors acting on the object have both a direction and a magnitude

Component vectors are sometimes drawn with a dotted line and a **subscript** indicating horizontal or vertical
For example, F is the vertical component of the force F

v

Calculating Vectors Graphically

Vectors at right angles to one another can be **combined** into one **resultant vector** The resultant vector will have the same effect as the two original ones

To calculate vectors graphically means carefully producing a scale drawing with all lengths and angles correct
This should be done using a sharp **pencil**, **ruler** and **protractor**

Follow these steps to carry out calculations with vectors on graphs

1. Choose a **scale** which fits the page
For example, use 1 cm = 10 m or 1 cm = 1 N, so that the diagram is around 10 cm high
2. Draw the vectors at **right angles** to one another
3. Complete the rectangle
4. Draw the **resultant vector** diagonally from the origin
5. Carefully **measure the length** of the resultant vector
6. Use the scale factor to calculate the **magnitude**
7. Use the protractor to measure the **angle**

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Vectors can be measured or calculated graphically if you are confident in using scales

Combining Vectors by Calculation

- In this method, a diagram is still essential but it does **not** need to be exactly to scale
- The diagram can take the form of a sketch, as long as the resultant, component and sides are clearly labelled

Resolving two force vectors F_1 and F_2 into a resultant force vector F_R

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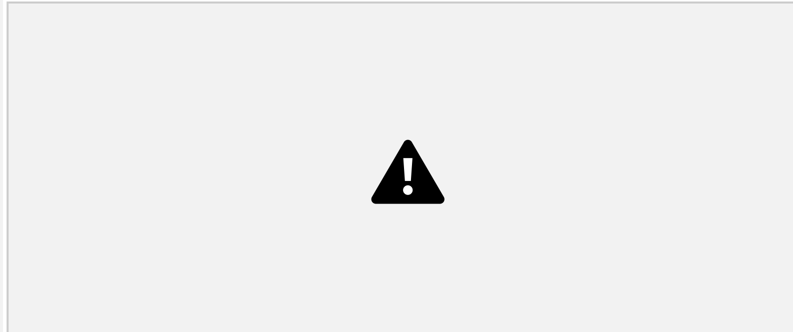
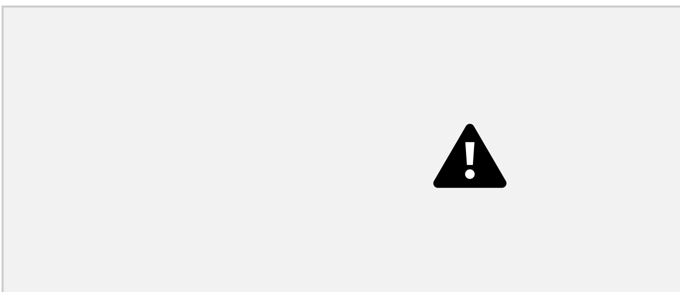
Use **trigonometry** to find the angle

The mnemonic **'soh-cah-toa'** is used to remember how to apply sines and cosines to resolve the sides of a triangle



Use **Pythagoras' Theorem** to find the resultant vector



Pythagoras's Theorem makes calculating vectors at right angles much simpler

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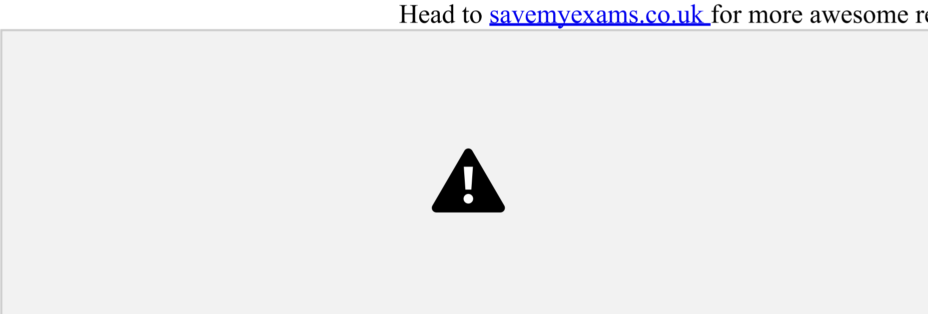


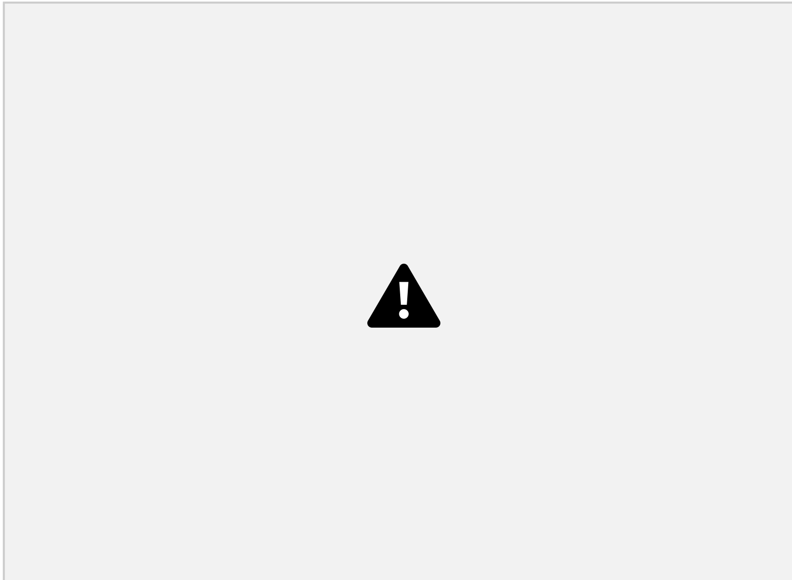
Trigonometry and Pythagoras' Therom are essential in vector calculations

Worked Example

A hiker walks a distance of 6 km due east and 10 km due north.

Calculate the magnitude of their displacement and its direction from the horizontal.

Step 1: Draw a vector diagram



Step 2: Calculate the magnitude of the resultant vector using Pythagoras' Theorem

$$\text{Resultant vector} = 6^2 + 10^2$$

$$\text{Resultant vector} = 136$$

Resultant vector = **11.66**

Step 3: Calculate the direction of the resultant vector using trigonometry

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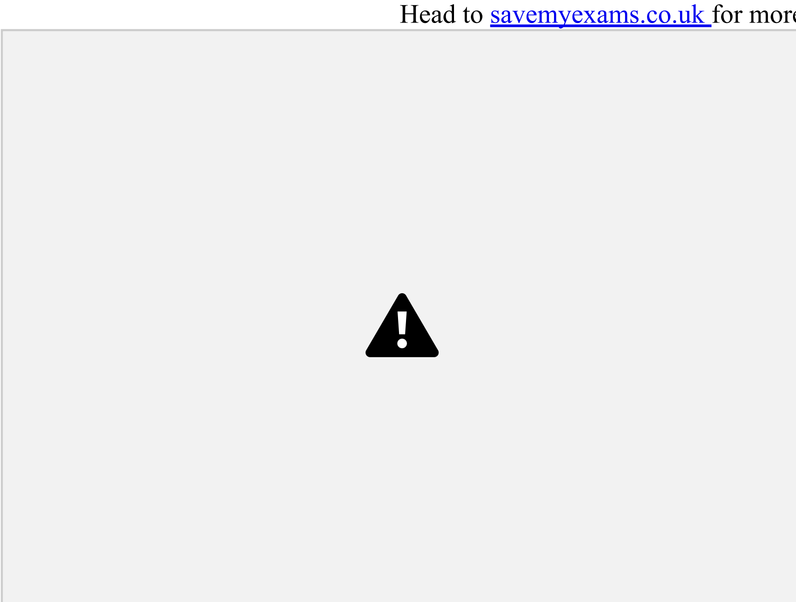

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$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{10}{6}$$

$$\theta = \tan^{-1}\left(\frac{10}{6}\right) = 59°$$

Step 4: State the final answer complete with direction

Resultant vector = **12 km 59° east and upwards from the horizontal**

Exam Tip

If the question specifically asks you to use the calculation or graphical method, you must solve the problem as asked. However, if the choice is left up to you then any correct method will lead to the correct answer.

The graphical method sometimes feels easier than calculating, but once you are confident with trigonometry and Pythagoras you will find calculating quicker and more accurate.

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Speed

- The **speed** of an object is the distance it travels per unit time
- Speed is a **scalar** quantity
  - This is because it only contains a magnitude (without a direction)
- For objects that are moving with a constant speed, use the equation below to calculate the speed:

Where:

- Speed is measured in metres per second (m/s)
- Distance travelled is measured in metres (m)
- Time taken is measured in seconds (s)

A hiker might have an average speed of 2.0 m/s, whereas a particularly excited bumble bee can have average speeds of up to 4.5 m/s

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Average Speed

In some cases, the speed of a moving object is not constant
For example, the object might be moving faster or slower at certain moments in time (accelerating and decelerating)

The equation for calculating the **average speed** of an object is:

$$\text{Average speed} = \frac{\text{distance travelled}}{\text{time taken}}$$

The formula for **average speed** (and the formula for **speed**) can be rearranged with the help of the formula triangle below:

How to Use Formula Triangles

Formula triangles are really useful for knowing how to rearrange physics equations To use them:

1. Cover up the quantity to be calculated, this is known as the 'subject' of the equation

2. Look at the position of the other two quantities
If they are on the same line, this means they are **multiplied**
If one quantity is above the other, this means they are **divided** - make sure to keep the order of which is on the top and bottom of the fraction!

In the example below, to calculate speed, cover-up 'speed' and only distance and time are left
This means it is equal to distance (on the top) ÷ time (on the bottom)

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Worked Example

How to use formula triangles

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Planes fly at typical speeds of around 250 m/s. Calculate the total distance travelled by a plane moving at this average speed for 2 hours.

Step 1: List the known quantities

Average speed = 250 m/s
Time taken = 2 hours

Step 2: Write the relevant equation

$$\text{Average speed} = \frac{\text{distance travelled}}{\text{time taken}}$$

Step 3: Rearrange for the total distance

total distance = average speed × time taken

Step 4: Convert any units

The time given in the question is not in standard units
Convert 2 hours into seconds:

2 hours = 2 × 60 × 60 = 7200 s

Step 5: Substitute the values for average speed and time taken total

distance = 250 × 7200 = 1 800 000 m

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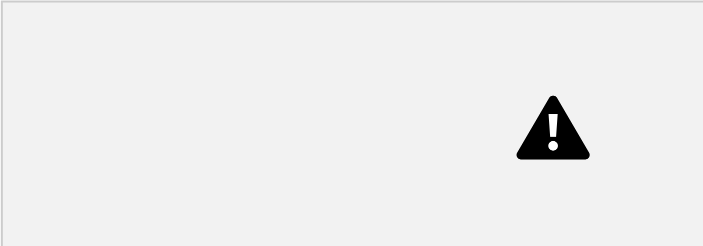
Velocity

The **velocity** of a moving object is similar to its speed, except it also describes the object's **direction**

The speed of an object only contains a magnitude - it's a **scalar** quantity The velocity of an object contains both **magnitude and direction**, e.g. '15 m/s south' or '250 mph on a bearing of 030°'

Velocity is therefore a **vector** quantity because it describes both magnitude and direction



The equation for velocity is very similar to the equation for speed:

$$\text{velocity} = \frac{\text{displacement}}{\text{time}}$$

$$v = \frac{s}{t}$$

Where:

v = velocity in metres per second (m/s)

s = displacement, measured in metres (m)

t = time, measured in seconds (s)

Velocity is a vector quantity, so it uses displacement, s, rather than distance which is scalar.

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Acceleration

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Acceleration is defined as the **rate of change of velocity**

- In other words, it describes how much an object's velocity **changes** every **second**

The equation below is used to calculate the average acceleration of an object:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{change in time}}$$

$$a = \frac{\Delta v}{\Delta t}$$

Where:

- a = acceleration in metres per second squared (m/s^2)
- Δv = change in velocity in metres per second (m/s)
- Δt = time taken in seconds (s)

The **change in velocity** is found by the **difference** between the initial and final velocity, as written below:

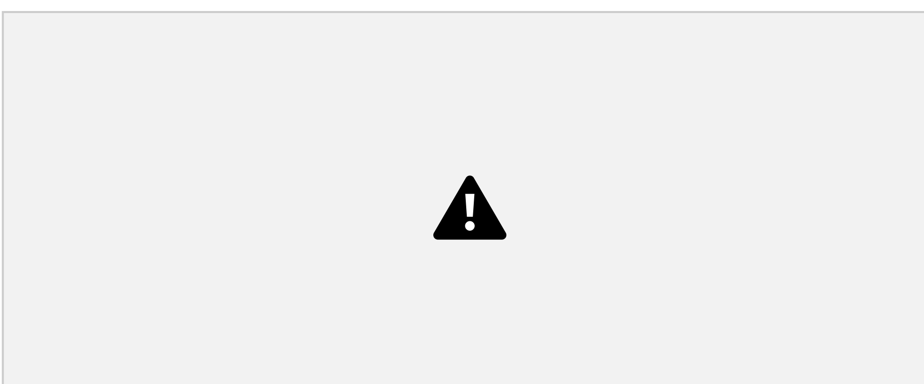
$$\text{change in velocity} = \text{final velocity} - \text{initial velocity}$$

$$\Delta v = v - u$$

Where:

- v = final velocity in metres per second (m/s)
- u = initial velocity in metres per second (m/s)

The equation for acceleration can be rearranged with the help of a formula triangle as shown:



Speeding Up & Slowing Down

- An object that speeds up is **accelerating**

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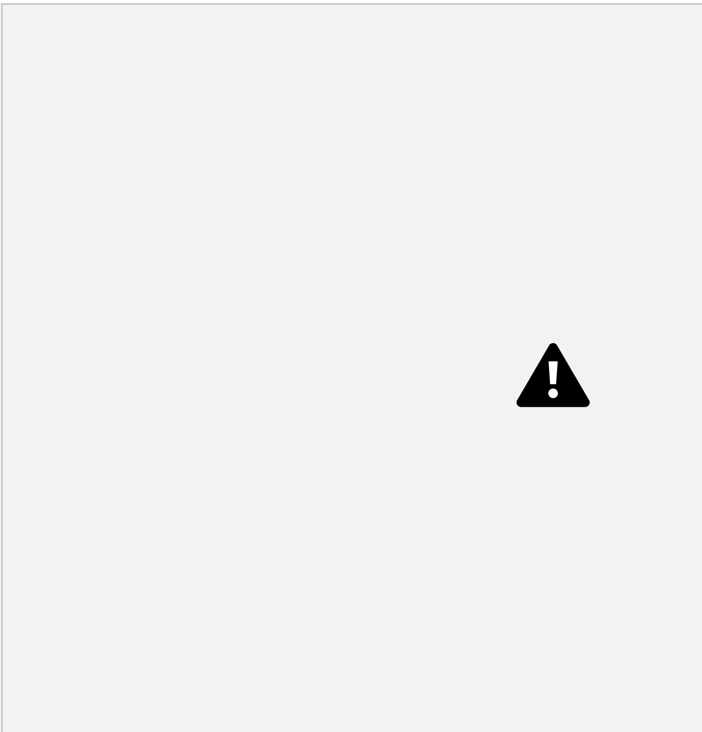
An object that slows down is **decelerating**

The **acceleration** of an object can be **positive** or **negative**, depending on whether the object is speeding up or slowing down

If an object is **speeding up**, its acceleration is **positive**

If an object is **slowing down**, its acceleration is **negative** (sometimes called **deceleration**)



A rocket speeding up (accelerating) and a car slowing down (decelerating)

Worked Example

A Japanese bullet train decelerates at a constant rate in a straight line. The velocity of the train decreases from 50 m/s to 42 m/s in 30 seconds.

(a) Calculate the change in velocity of the train.

(b) Calculate the deceleration of the train, and explain how your answer shows the train is slowing down.

Part (a)

Step 1: List the known quantities

Initial velocity = 50 m/s
Final velocity = 42 m/s

Step 2: Write the relevant equation

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Step 3: Substitute values for final and initial velocity

change in velocity = final velocity − initial velocity change in velocity = 42 − 50 = −8 m/s

Part (b)

Step 1: List the known quantities

Change in velocity, $\Delta v = -8$ m/s
Time taken, $t = 30$ s

Step 2: Write the relevant equation

$$a = \frac{\Delta v}{\Delta t}$$

Step 3: Substitute the values for change in velocity and time $a = -8 \div 30 = -0.27$ m/s

Step 4: Interpret the value for deceleration

The answer is **negative**, which indicates the train is **slowing down**

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Exam Tip

Remember the units for acceleration are **metres per second squared**, m/s . In other words, acceleration measures how much the velocity (in m/s) changes every second, m/s/s.

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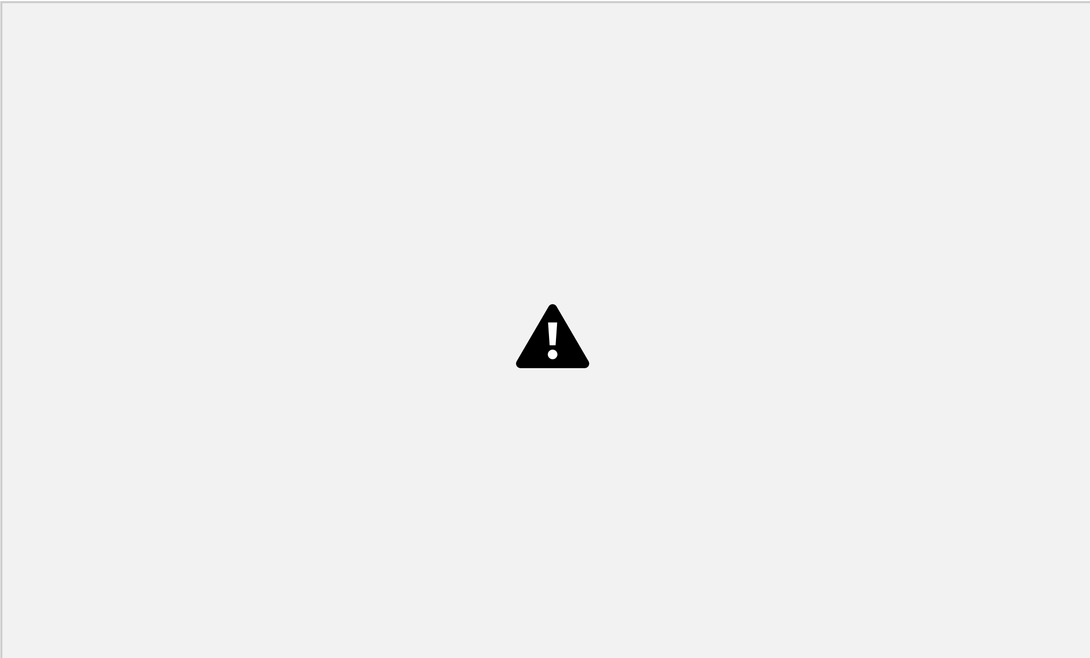
Distance-Time Graphs

A distance-time graph shows how the **distance** of an object moving in a straight line (from a starting position) varies over time:



This graph shows a moving object moving further away from its origin

Constant Speed on a Distance-Time Graph

Distance-time graphs also show the following information:

- If the object is moving at a **constant speed**
- How **large** or **small** the speed is

A **straight line** represents **constant speed**

The slope of the straight line represents the **magnitude** of the speed:

- A very **steep** slope means the object is moving at a **large** speed
- A **shallow** slope means the object is moving at a **small** speed
- A **flat**, **horizontal line** means the object is **stationary** (not moving)

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This graph shows how the slope of a line is used to interpret the speed of moving objects. Both of these objects are moving with a constant speed, because the lines are straight.

Changing Speed on a Distance-Time Graph

- Objects might be moving at a **changing speed**
  - This is represented by a **curve**
- In this case, the slope of the line will be changing
  - If the slope is **increasing**, the **speed** is **increasing** (accelerating)
  - If the slope is **decreasing**, the **speed** is **decreasing** (decelerating)
- The image below shows two different objects moving with changing speeds

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Changing speeds are represented by changing slopes. The red line represents an object slowing down and the green line represents an object speeding up.

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Using Distance-Time Graphs

The **speed** of a moving object can be calculated from the **gradient** of the line on a **distance-time** graph:

The speed of an object can be found by calculating the gradient of a distance-time graph

The **rise** is the **change** in y (distance) values
The **run** is the **change** in x (time) values

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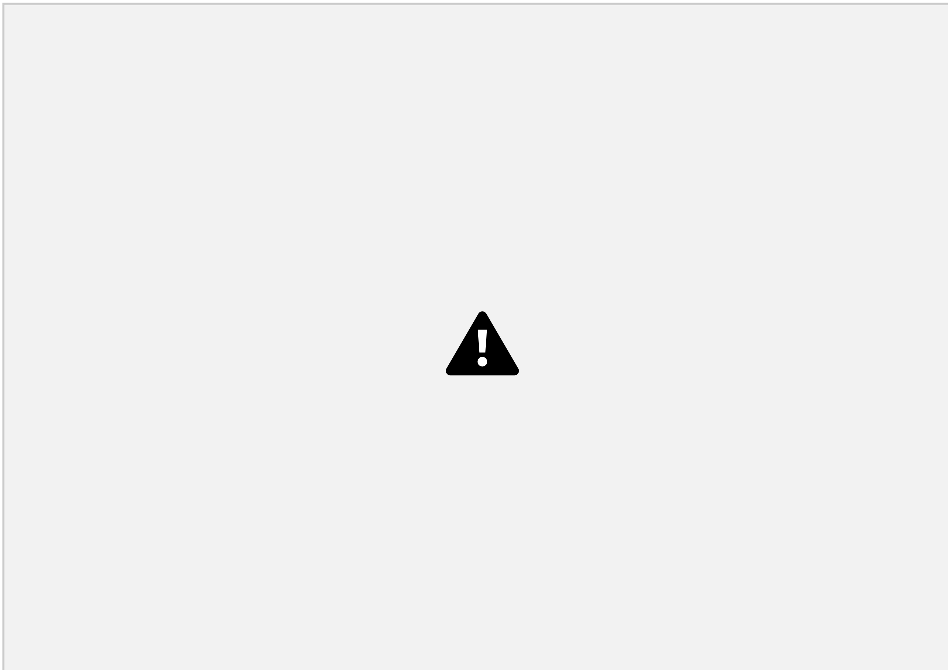
Worked Example

A distance-time graph is drawn below for part of a train journey. The train is travelling at a constant speed.



Calculate the speed of the train.

Step 1: Draw a large gradient triangle on the graph and label the magnitude of the rise and run

The image below shows a large **gradient triangle** drawn with dashed lines
The **rise** and **run** magnitude is labelled, using the **units** as stated on each axes

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Step 2: Convert units for distance and time into standard units

The distance travelled (rise) = 8 km = **8000 m**
The time taken (run) = 6 mins = **360 s**

Step 3: State that speed is equal to the gradient of a distance-time graph

The **gradient** of a **distance-time** graph is equal to the **speed** of a moving object:

Step 4: Substitute values in to calculate the speed

speed = gradient = 8000 ÷ 360

speed = **22.2 m/s**

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Worked Example

Ose decides to take a stroll to the park. He finds a bench in a quiet spot and takes a seat, picking up where he left off reading his book on Black Holes.
After some time reading, Ose realises he lost track of time and runs home.

A distance-time graph for his trip is drawn below.

a)
How long does Ose spend reading his book?
b)
There are three sections labelled on the graph, A, B and C. Which section represents Ose running home?
c)
What is the total distance travelled by Ose?

Part (a)

Ose spends **40 minutes** reading his book
The **flat** section of the line (section B) represents an object which is **stationary** - so section B represents Ose sitting on the bench reading
This section lasts for **40 minutes** - as shown in the graph below

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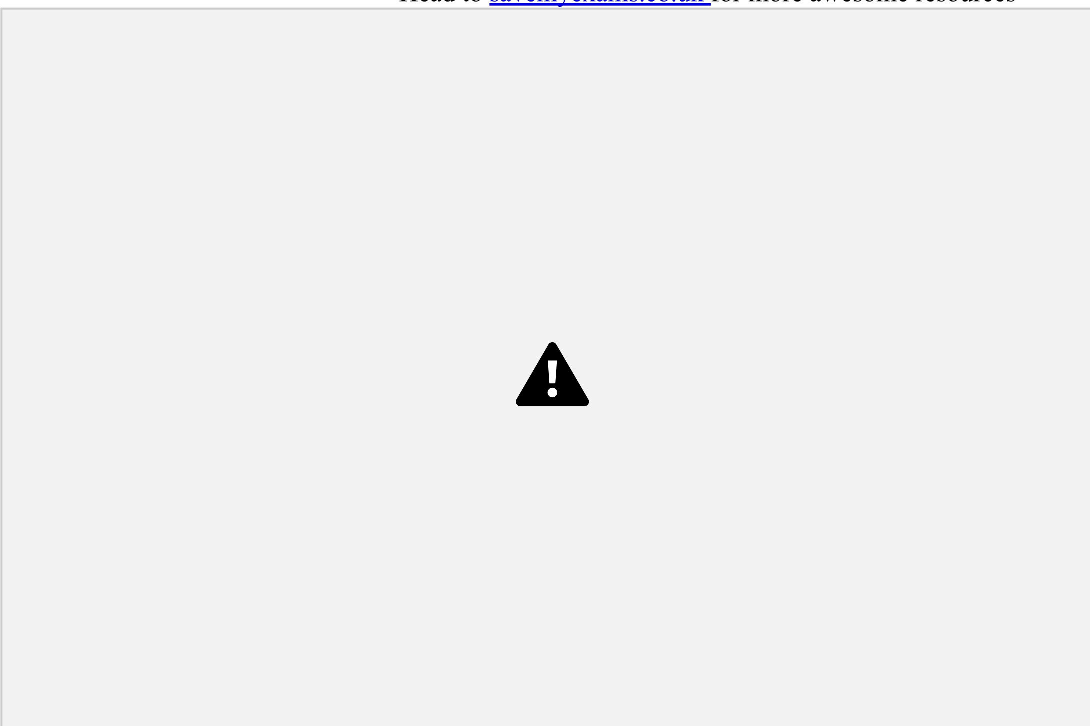

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Part (b)

Section C represents Ose running home
The **slope** of the line in section C is **steeper** than the slope in section A
This means Ose was moving with a **larger** speed (running) in section C

Part (c)

The total distance travelled by Ose is **0.6 km**
The total **distance** travelled by an object is given by the final point on the line - in this case, the line ends at **0.6 km** on the **distance** axis. This is shown in the image below:

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Exam Tip

Use the **entire line**, where possible, to calculate the gradient. Examiners tend to award credit if they see a **large gradient triangle** used - so remember to draw these directly on the graph itself!

Remember to check the **units** of variables measured on each axis. These may not always be in standard units - in our example, the unit of distance was **km** and the unit of time was **minutes**. Double-check which units to use in your answer.

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Speed-Time Graphs

A speed-time graph shows how the speed of a moving object varies with time

- The red line represents an object with **increasing** speed
- The green line represents an object with **decreasing** speed

Increasing and decreasing speed represented on a speed-time graph

Acceleration on a Speed-Time Graph

Speed-time graphs also show the following information:

- If the object is moving with a **constant** acceleration or deceleration
- The **magnitude** of the acceleration or deceleration

- A **straight line** represents **constant acceleration**
- The **slope** of the line represents the **magnitude** of acceleration
    - A **steep** slope means **large acceleration** (or deceleration) - i.e. the object's speed changes very **quickly**
    - A **gentle** slope means **small acceleration** (or deceleration) - i.e. the object's speed changes very **gradually**
    - A **flat line** means the acceleration is **zero** - i.e. the object is moving with a **constant speed**

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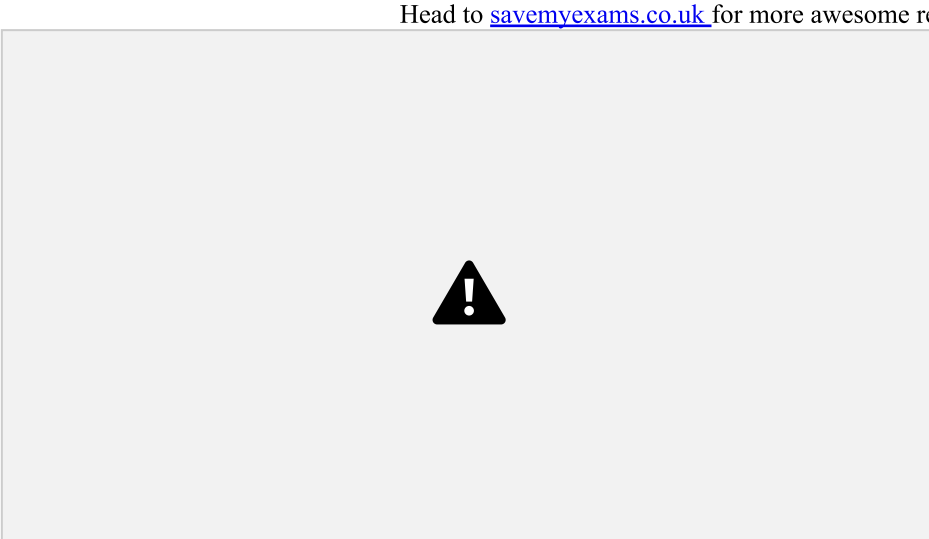

This image shows how to interpret the slope of a speed-time graph

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Using Speed-Time Graphs

The **distance travelled** by an object can be found by determining the **area beneath the graph**

The distance travelled can be found from the area beneath the graph

If the area beneath the graph forms a **triangle** (the object is accelerating or decelerating) then the area can be determined using the formula:

$$\text{area} = \frac{1}{2} \times \text{base} \times \text{height}$$

If the area beneath the graph is a **rectangle** (constant velocity) then the area can be determined using the formula:

$$\text{area} = \text{base} \times \text{height}$$

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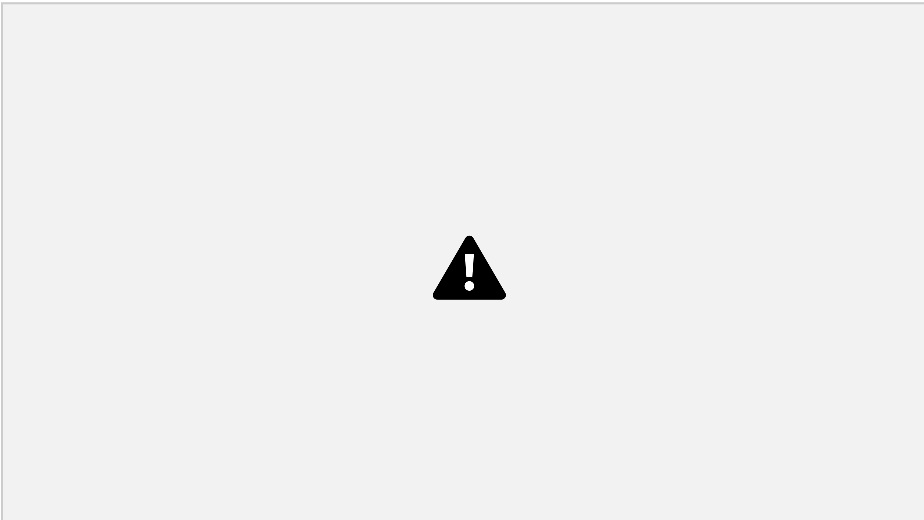
Worked Example

The speed-time graph below shows a car journey which lasts for 160 seconds.



Calculate the total distance travelled by the car on this journey.

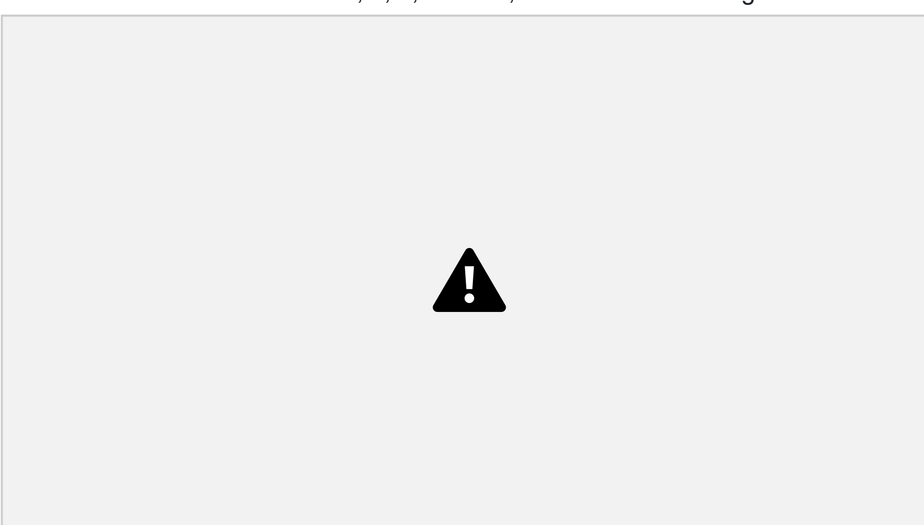
Step 1: Recall that the area under a velocity-time graph represents the distance travelled

In order to calculate the total distance travelled, the total area underneath the line must be determined

Step 2: Identify each enclosed area

In this example, there are **five** enclosed areas under the line

These can be labelled as areas 1, 2, 3, 4 and 5, as shown in the image below:



Step 3: Calculate the area of each enclosed shape under the line

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Area 1 = area of a triangle = ½ × base × height = ½ × 40 × 17.5 = **350 m** Area 2 = area of a rectangle = base × height = 30 × 17.5 = **525 m** Area 3 = area of a triangle = ½ × base × height = ½ × 20 × 7.5 = **75 m** Area 4 = area of a rectangle = base × height = 20 × 17.5 = **350 m** Area 5 = area of a triangle = ½ × base × height = ½ × 70 × 25 = **875 m**

Step 4: Calculate the total distance travelled by finding the total area under the line

Add up each of the five areas enclosed:

total distance = 350 + 525 + 75 + 350 + 875

total distance = **2175 m**

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Interpreting Speed-Time graphs

EXTENDED

- If there is a **change in** an object's **speed**, then it is **accelerating**

- An object may accelerate at a steady rate, this is called **constant acceleration**
    - On a speed-time graph this will be a non-horizontal **straight line**

- An object may accelerate at an **increasing rate**
    - On a speed-time graph this would be an **upward curve**

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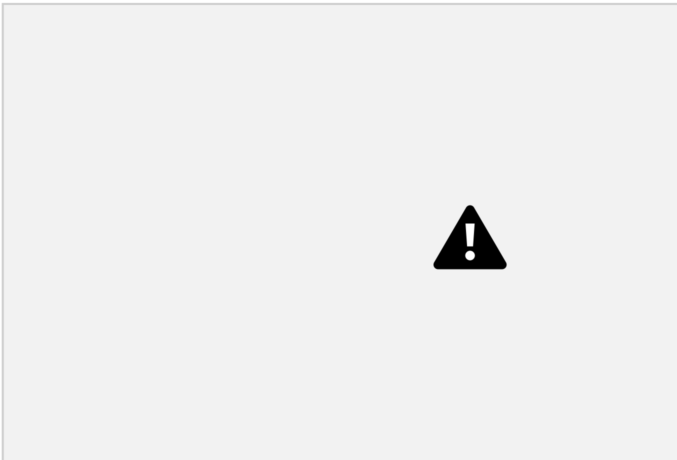
An object may accelerate at a **decreasing rate**
On a speed-time graph this would be an **downward curve**

Calculating Acceleration

The **acceleration** of an object can be calculated from the **gradient** of a speed-time graph

acceleration = gradient = rise

run



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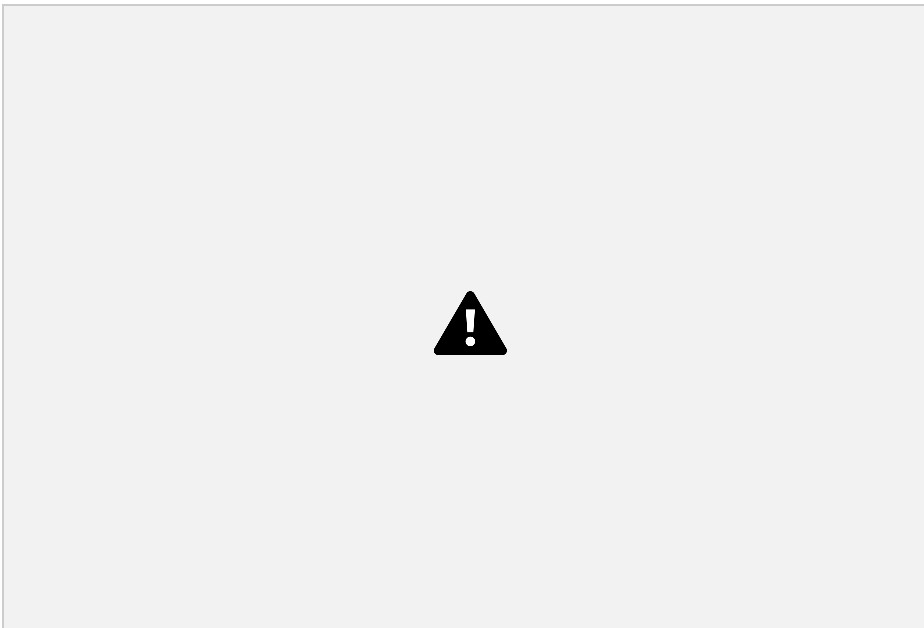
Worked Example

Tora is training for a cycling tournament.

The speed-time graph below shows her motion as she cycles along a flat, straight road.



(a) In which section (A, B, C, D, or E) of the speed-time graph is Tora's acceleration the largest?

(b) Calculate Tora's acceleration between 5 and 10 seconds.

Part (a)

Step 1: Recall that the slope of a speed-time graph represents the magnitude of acceleration

The slope of a speed-time graph indicates the magnitude of acceleration
Therefore, the only sections of the graph where Tora is accelerating is section B and section D

Sections A, C, and E are flat – in other words, Tora is moving at a constant speed (i.e. not accelerating)

Step 2: Identify the section with the steepest slope

Section D of the graph has the steepest slope

Hence, the largest acceleration is shown in section D

Part (b)

Step 1: Recall that the gradient of a speed-time graph gives the acceleration

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Calculating the gradient of a slope on a speed-time graph gives the acceleration for that time period

Step 2: Draw a large gradient triangle at the appropriate section of the graph

A gradient triangle is drawn for the time period between 5 and 10 seconds below:

Step 3: Calculate the size of the gradient and state this as the acceleration

The acceleration is given by the gradient, which can be calculated using:

acceleration = gradient = 5 ÷ 5 = 1 m/s^2

Therefore, Tora accelerated at 1 m/s^2 between 5 and 10 seconds

Exam Tip

Use the **entire slope**, where possible, to calculate the gradient. Examiners tend to award credit if they see a **large gradient triangle** used - so remember to draw 'rise' and 'run' lines directly on the graph itself!

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Acceleration of Free Fall

In the absence of **air resistance**, all objects fall with the **same acceleration** This is called the **acceleration of freefall** (this is also sometimes called acceleration due to gravity)

$$\text{acceleration of freefall} = g = 9.81\ \text{m/s}^2$$

In the absence of air resistance, Galileo discovered that all objects (near Earth's surface) fall with an acceleration of about 9.8 m/s^2

This means that for every second an object falls, its velocity will increase by 9.8 m/s

The symbol **g** also stands for the **gravitational field strength**, and can be used to calculate the weight of an object using its mass:

$$\text{weight} = \text{mass} \times \text{gravitational field strength}$$

$$W = mg$$

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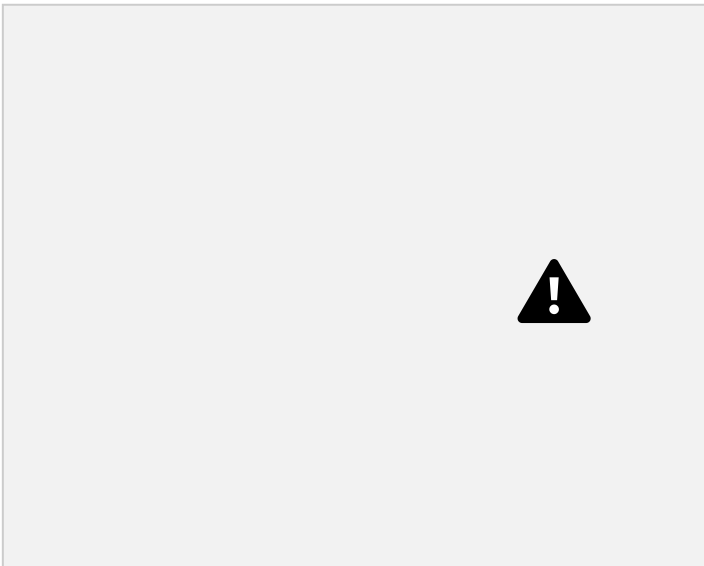
Motion of Falling Objects

EXTENDED

Falling Objects without Air Resistance

In the **absence of air resistance**, all objects falling in a uniform gravitational field, fall with the **same acceleration**, regardless of their mass

So long as air resistance remains insignificant, the speed of a falling object will **increase** at a **steady rate**, getting larger the longer it falls for.



In the absence of air resistance objects fall with constant acceleration

Falling Objects with Air Resistance

Objects falling through fluids (fluids are liquids or gases) in a uniform gravitational field, experience **two forces**:
Weight (due to gravity)
Friction (such as air resistance)

A skydiver jumping from a plane will experience:
A downward acting force of weight (mass × acceleration of freefall) An upward acting force of air resistance (frictional forces always oppose the direction of motion)

The force of air resistance increases with speed. This is illustrated in the image below:

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Debbie initially accelerates downwards due to her weight. The upwards air resistance increases as she falls until it eventually grows big enough to balance the weight force

- Initially, the upwards air resistance is **very small** because the skydiver isn't falling very quickly
    - Therefore, there are **unbalanced** forces on the skydiver initially
- As the skydiver speeds up, air resistance **increases,** eventually growing large enough to **balance the downwards weight force**
- Once **air resistance equals weight,** the forces are **balanced**
    - This means there is no longer any **resultant force**
- Therefore, the skydiver's acceleration is **zero** - they now travel at a **constant** speed
- This speed is called their **terminal velocity**
- When the skydiver opens the parachute, the **air resistance increases**
    - This is due to the increased surface area of the parachute opening
- The upward force of air resistance on the skydiver increases, **slowing the acceleration** of the skydivers fall
    - The skydiver **decelerates**
- Eventually, the **forces balance** out again, and a new **slower terminal velocity** is reached

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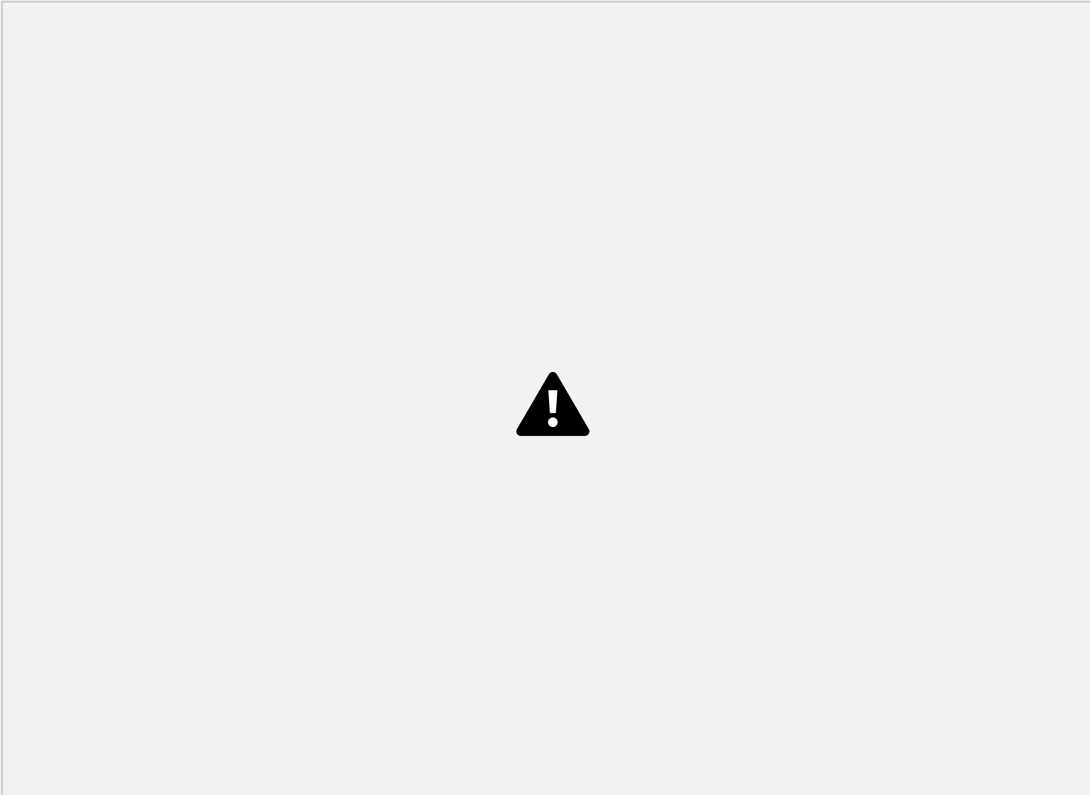

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Graph showing how the velocity of a skydiver changes during the descent

Worked Example

A small object falls out of an aircraft. Choose words from the list to complete the sentences below:

Friction Gravitational field strength Air pressure

Accelerates Falls at a steady speed Slows down

(a) The weight of an object is the product of the object's mass and the ___________.

(b) When an object falls, initially it _____________.

(c) As the object falls faster, the force of _______________ acting upon the object increases.

(d) Eventually the object _______________ when the force of friction equals the force of weight acting on it.

Part (a)

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The weight of an object is the product of the object's mass and the **gravitational field strength**.

The weight force is due to the Earth's gravitational pull on the object's mass as it falls through a uniform gravitational field

Part (b)

When an object falls, initially it **accelerates.**

The **resultant force** on the object is **very large** initially, so it accelerates This is because there is a **large unbalanced force** downwards (its weight) - the upward force of air resistance is very small to begin with

Part (c)

As the object falls faster, the force of **friction** acting upon the object increases.

The force of **air resistance** is due to **friction** between the object's motion and **collisions with air particles**
Collisions with air particles slow the object down, so air itself produces a **frictional force**, called air resistance (sometimes called **drag**)

Part (d)

Eventually the object **falls at a steady speed** when the force of friction equals the force of weight acting on it.

When the upwards **air resistance** increases enough to **balance** the downwards **weight** force, the resultant force on the object is **zero**
This means the object isn't accelerating - rather, it is moving at a **steady** (terminal) **speed**

Exam Tip

The force of gravity on an object with mass is called **weight**. If asked to name this force make sure you use this word: Don’t refer to it as “gravity” as this term could also mean gravitational field strength and so would probably be marked wrong.

Likewise, remember to identify **air resistance** as the upwards force on a falling object. This force **gets larger** as the object speeds up, but the weight of the object stays constant. Don't confuse 'air resistance' with 'air pressure' - these are two different concepts!

Exam questions about terminal velocity tend to involve the motion of skydivers as they fall

A common misconception is that skydivers move upwards when their parachutes are deployed - however, this is not the case, they are in fact **decelerating** to a lower terminal velocity

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Mass

Mass is a measure of the quantity of **matter** in an object at rest relative to the observer
- Mass is a **scalar** quantity
- The SI unit for mass is the **kilogram** (kg)

Consequently, mass is the property of an object that resists change in motion The greater the mass of an object, the more difficult it is to speed it up, slow it down, or change its direction

A mass may sometimes be given in **grams** (g)
- 1000 g = 1 kg
- 1 g = 0.001 kg

Worked Example

A mass is 2.7 kg.

State the number of grams in 2.7 kg.

1. State the conversion between g and kg

1 kg = 1000 g

2. Convert 2.7 kg into g

2.7 kg = 2.7 × 1000 = 2700 g

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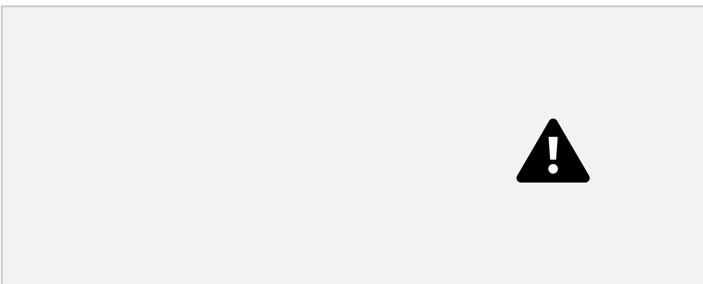
Weight

Weight is a gravitational **force** on an object with mass
Weight is a force, so it is a **vector** quantity
The SI units for force are **newtons** (N)

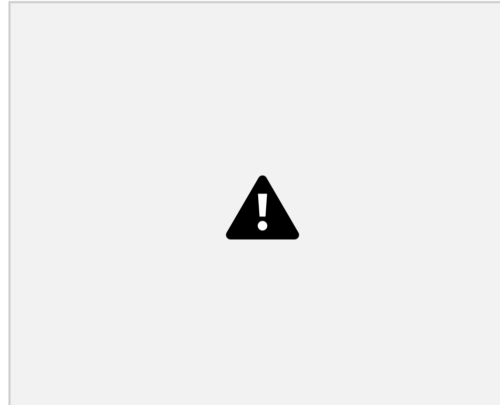
EXTENDED

Weight is the effect of a **gravitational field** on a mass

The weight of a body is equal to the product of its mass (m) and the acceleration of free fall (g)



You can rearrange this equation with the help of the formula triangle:



Use the formula triangle to help you rearrange the equation

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Gravitational Field Strength

Gravitational field strength is defined as:

The force per unit mass acting on an object in a

gravitational field On Earth, this is equal to **9.8 N/kg**

Gravitational field strength is also known as acceleration of free fall, or acceleration due to gravity

In this context the units are m/s^2

The value of g (gravitational field strength) varies from planet to planet depending on their mass and radius

A few examples of varying gravitational field strength are shown below:

Gravitational field strength of the planets in our solar system

Mass v Weight

An object's mass always remains the same, however, its weight will differ depending on the strength of the gravitational field on different planets For example, the gravitational field strength on the Moon is **1.63 N/kg**, meaning an object's weight will be about **6 times** less than on Earth

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On the moon, your mass will stay the same but your weight will be much lower

Exam Tip

You won't be expected the learn the exact value of g (9.81 N/kg), but you will be expected to remember that g = 10 N/kg and use it in calculations

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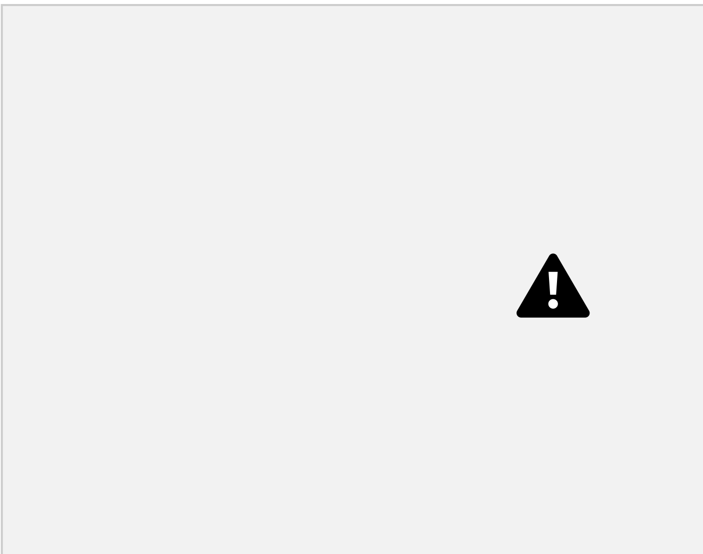
A balance can be used to compare two different weights

Using a Balance

The weight of two objects can be compared using a balance

Because the gravitational field strength is constant everywhere on Earth, this also allows us to measure the mass of an object

$$m = \frac{W}{g}$$



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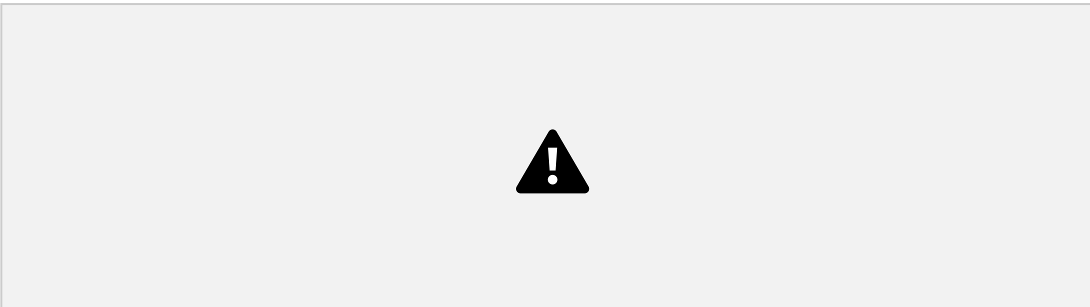
Density

Density is defined as:

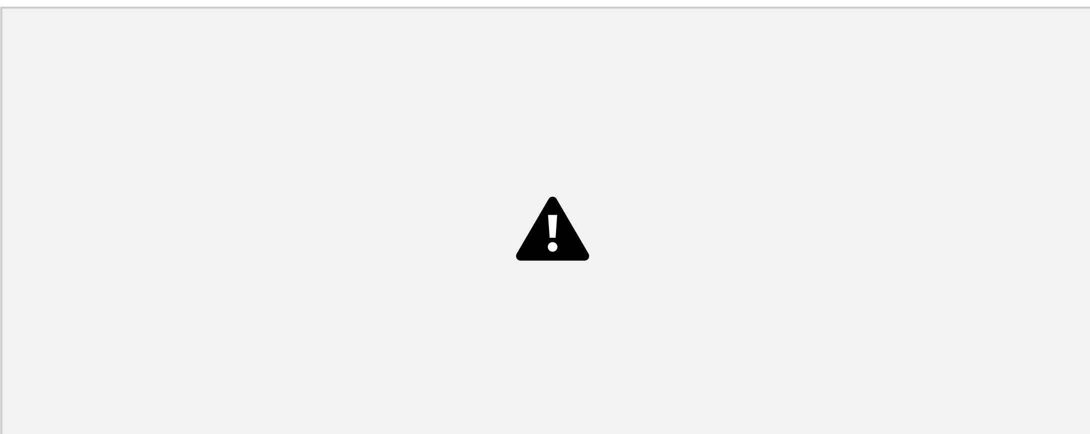
The mass per unit volume of a material

- Objects made from **low density** materials typically have a **low mass**
- Similarly sized objects made from **high density** materials have a **high mass**
  - For example, a bag full of feathers is far lighter compared to a similar bag full of metal
  - Or another example, a balloon is less dense than a small bar of lead despite occupying a larger volume

Density is related to mass and volume by the following equation:



- Gases, for examples, are less dense than solids because the molecules are more spread out (same mass, over a larger volume)



Gases are less dense than solids

- This equation can be rearranged with the help of the formula triangle:

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Density, mass, volume formula triangle

The units of density depend on what units are used for mass and volume:

- If the mass is measured in g and volume in cm^3, then the density will be in g/cm^3
- If the mass is measured in kg and volume in m^3, then the density will be in kg/m^3

This table gives some examples of densities on common materials

- If a material is more dense than water ($1000 kg/m^3$), then it will sink

Approximate Densities of Materials Table

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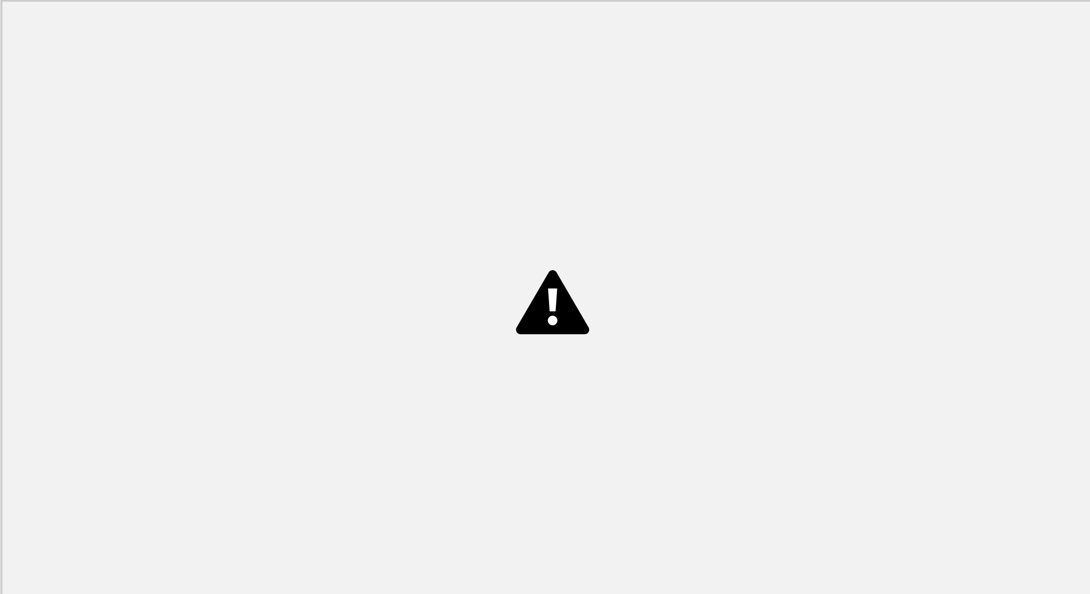
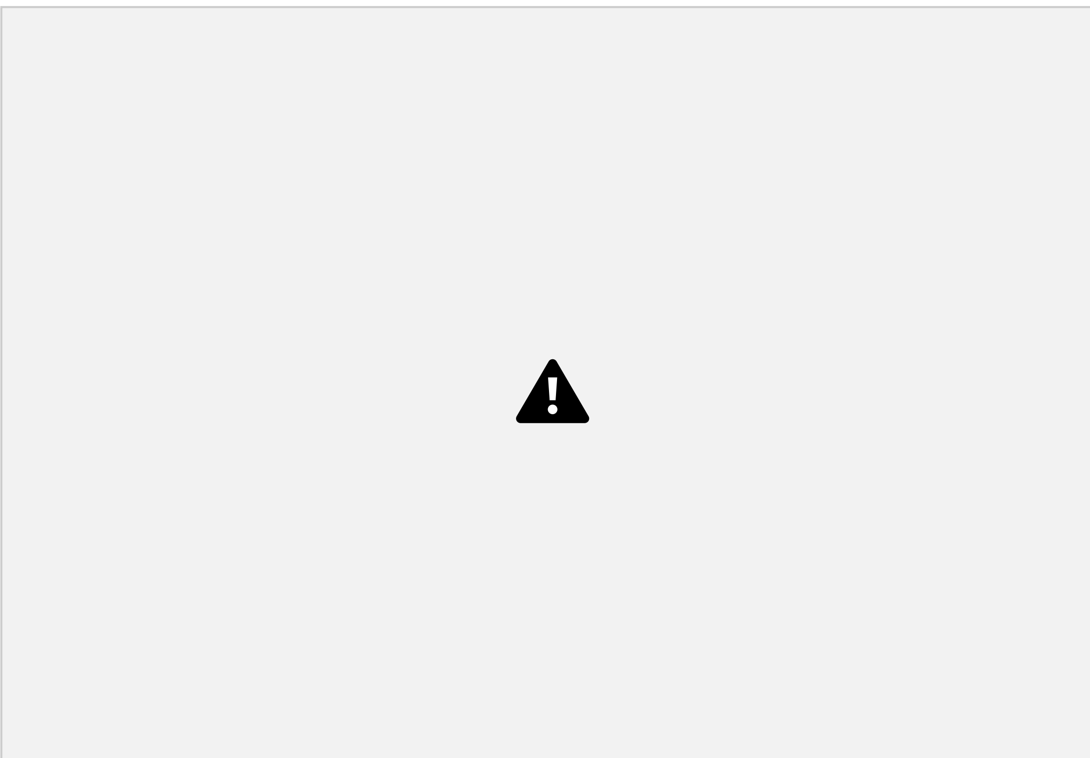


The volume of an object may not always be given directly, but can be calculated with the appropriate equation depending on the object's shape



Volumes of common 3D shapes

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Worked Example

A paving slab has a mass of 73 kg and dimensions 0.04 m × 0.5 m × 0.85 m.

Calculate the density, in kg/m^3, of the material from which the paving slab is made.

Step 1: List the known quantities

Mass of slab, m = 73 kg

Volume of slab, V = 0.04 m × 0.5 m × 0.85 m = 0.017 m^3

Step 2: Write out the equation for density

Step 3: Substitute in values

ρ = 73 ÷ 0.017 = 4294 kg/m^3

Step 4: Round the answer to two significant figures

3

ρ = 4300 kg/m

Exam Tip

Make sure you are comfortable converting between units such as metres (m) and centimetres (cm) or grams (g) and kilograms (kg).

When converting a **larger** unit to a **smaller** one, you **multiply (×)**
E.g. 125 m = 125 × 100 = 12 500 cm

When you convert a **smaller** unit to a **larger** one, you **divide (÷)**

E.g. 5 g = 5 ÷ 1000 = 0.005 or 5 × 10 kg

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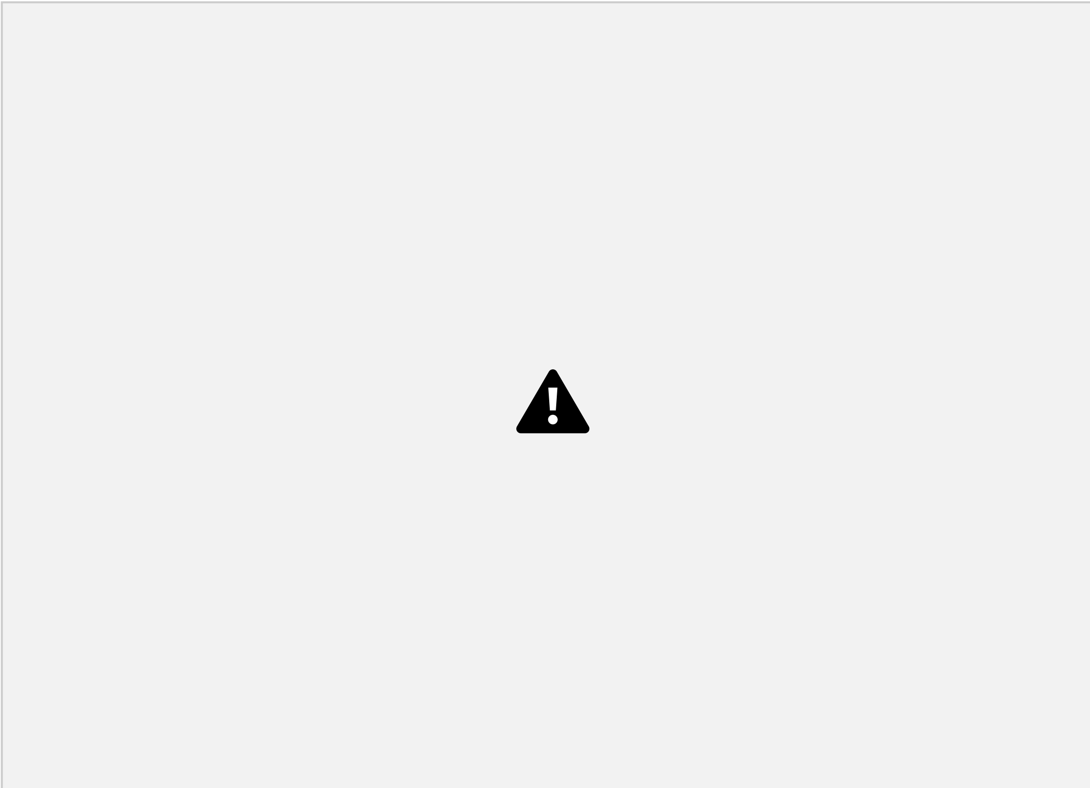
Measuring Density

Equipment List



Resolution of measuring equipment:

- 30 cm ruler = 1 mm
- Vernier calipers = 0.01 mm
- Micrometer = 0.001 mm
- Digital balance = 0.01 g

Experiment 1: Measuring the Density of Regularly Shaped Objects

The aim of this experiment is to determine the densities of regular objects by using measurements of their dimensions

Variables:

- Independent variable = Type of shape / volume
- Dependent variable = Mass of the object

Method

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1. Place the object on a digital balance and note down its mass
2. Use either the ruler, Vernier calipers or micrometer to measure the object's dimensions (width, height, length, radius) – the apparatus will depend on the size of the object
3. Repeat these measurements and take an average of these readings before calculating the density

An example of a results table might look like this:

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Analysis of Results

Calculate the volume of the object depending on whether it is a cube, sphere, cylinder (or other regular shape)

Calculating the volume of an object depends on its shape

Remember to convert from **centimetres (cm)** to **metres (m)** by dividing by 100 **1 cm = 0.01 m**

50 cm = 0.5 m

Using the mass and volume, the density of each can be calculated using the equation:

Where:

ρ = density in kilogram per metres cubed (kg/m^3)

m = mass in kilograms (kg)

V = volume in metres cubed (m^3)

Experiment 2: Measuring the Density of Irregularly Shaped Objects

The aim of this experiment is to determine the densities of irregular objects using a displacement technique

Variables:

Independent variable = Different irregular shapes / mass

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Method

Dependent variable = Volume of displaced water

Apparatus for measuring the density of irregular objects

1. Place the object on a digital balance and note down its mass
2. Fill the eureka can with water up to a point just below the spout 3. Place an empty measuring cylinder below its spout
4. Carefully lower the object into the eureka can
5. Measure the volume of the displaced water in the measuring cylinder 6. Repeat these measurements and take an average before calculating the density

Alternatively, the object can be placed in a measuring cylinder containing a known volume of liquid, and the change in volume then measured

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When an irregular solid is placed in a measuring cylinder, the level of the liquid will rise by an amount equal to the volume of the solid

Once the mass and volume of the shape is known, its density can be calculated

An example of a results table might look like this:

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Analysis of Results

- The volume of the water displaced is equal to the volume of the object
- Once the mass and volume of the shape are known, the density can be calculated using:

Experiment 3: Measuring Density of Liquids

- The aim of this experiment is to determine the density of a liquid by finding a difference in its mass

Variables:

- Independent variable = Volume of water added
- Dependent variable = Mass of cylinder

Method

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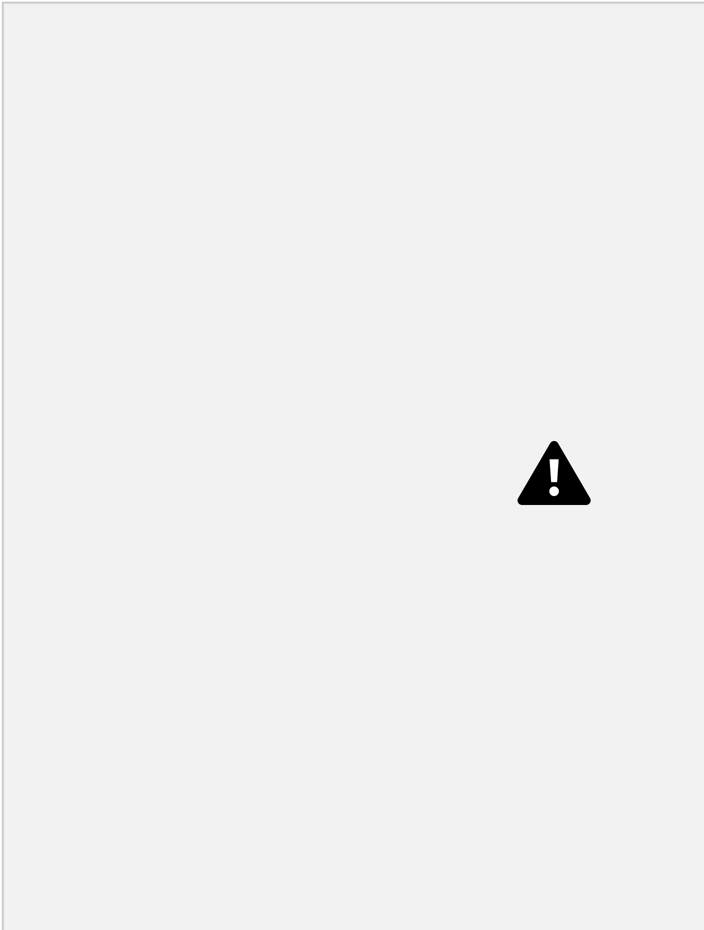
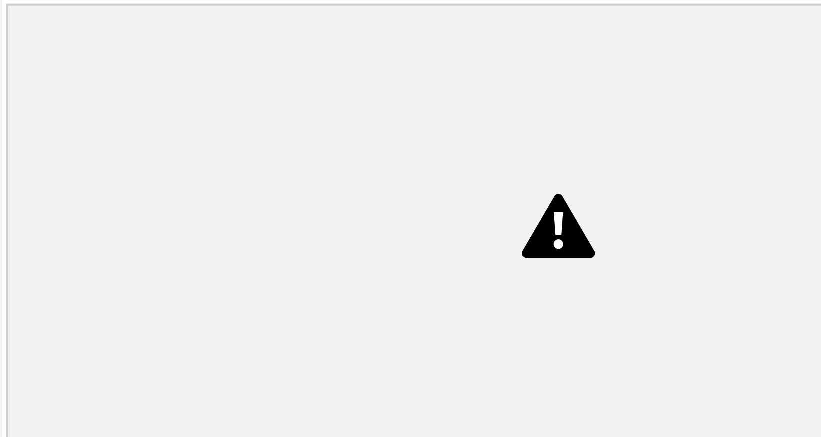


Apparatus for determining the density of a liquid

1. Place an empty measuring cylinder on a digital balance and note down the mass 2. Fill the cylinder with the liquid and note down the volume
3. Note down the new reading on the digital balance
4. Repeat these measurements and take an average before

calculating the density An example of a results table might look

like this:



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Analysis of Results

Find the mass of the liquid by subtracting the final reading from the original reading

Mass of liquid = Mass of cylinder with water – mass of

cylinder Remember to convert between **grams (g)** and

kilograms (kg) by dividing by 1000 **1 g = 0.001 kg**

78 g = 0.078 kg

Once the mass and volume of the liquid are known, the density can be calculated using the equation:

Evaluating the Experiments

Systematic Errors:

Ensure the digital balance is set to zero before taking measurements of mass This includes when measuring the density of the liquid – remove the measuring cylinder and zero the balance before adding the liquid

Random Errors:

A main cause of error in this experiment is in the measurements of length Ensure to take repeat readings and calculate an average to keep this error to a minimum

Place the irregular object in the displacement can carefully, as dropping it from a height might cause water to splash which will lead to an incorrect volume reading

Safety Considerations

There is a lot of glassware in this experiment, ensure this is handled carefully Water should not be poured into the measuring cylinder when it is on the electric balance
This could lead to electric shock

Make sure to stand up during the whole experiment, to react quickly to any spills

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Exam Tip

There is a lot of information to take in here! When writing about experiments, a good sequence is as follows:

- If you need to use an equation to calculate something, start off by giving it as this will give you some hints about what you need to mention later
- List the apparatus that you need
- State what measurements you need to make (your equation will give you some hints) and how you will measure them
- Finally, state that you will repeat each measurement several times and take averages

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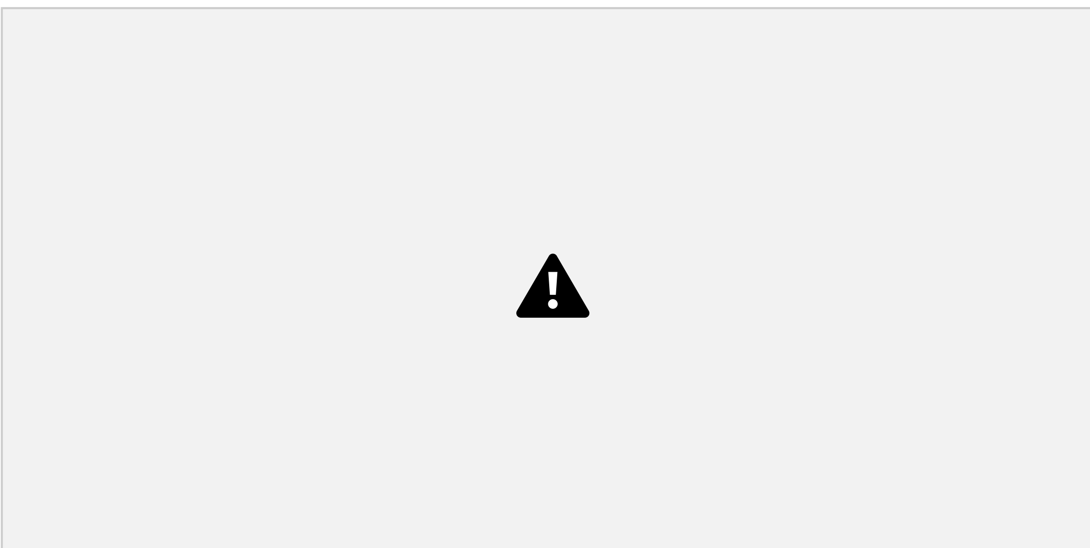
Floating Objects

Upthrust

- Upthrust is a force that pushes upwards on an object submerged in a **fluid** i.e. liquids and gases
- It is always in the opposite direction to the object's weight
    - This is why boats, and objects that are less dense than water, float

- The size of the upthrust depends on the **density** of the fluid as well as the **volume** of fluid that is displaced (which is equal to the volume of the object)
    - The **denser** the liquid, the **greater** the upthrust it will exert on an object



Upthrust is in the opposite direction to the weight of the boat and the fisherman

Factors Affecting Floating & Sinking

- Whether an object sinks or floats depends on the **upthrust**:
    - If the upthrust on an object is **equal** to (or **greater** than) the object's weight, then the object will **float**
    - If the upthrust is **smaller** than the weight then the object will sink

- The outcome also depends on the object's **density**:
    - If it has a density **less** than the density of the fluid it is immersed in, the object will **float**
    - If it has a density **more** than the density of the fluid it is immersed in, the object will **sink**

- This is because if the density of the object is greater than the density of the fluid, the object can never displace enough fluid to create an upthrust that will hold its weight up (and therefore sinks)

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Objects which are less dense than water will float and which are more dense will sink

- A polystyrene block will **float** in water
  - This is because polystyrene has a density of 0.05 g/cm^3 which is **much less** than the density of water (1.0 g/cm^3)
- A wooden block will be partially submerged but will still **float**
  - This is because the density of a wooden block (0.9 g/cm^3) is **slightly less** than the density of water
- An iron block will **sink**
  - This is because iron has a density (7.9 g/cm^3) that is **much higher** than water

Exam Tip

- The main thing to remember is that **density is mass per unit volume**
- In Physics, mass is almost always measured in kg
- Density is the only topic in which physicists sometimes use grams instead

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Floating Liquids

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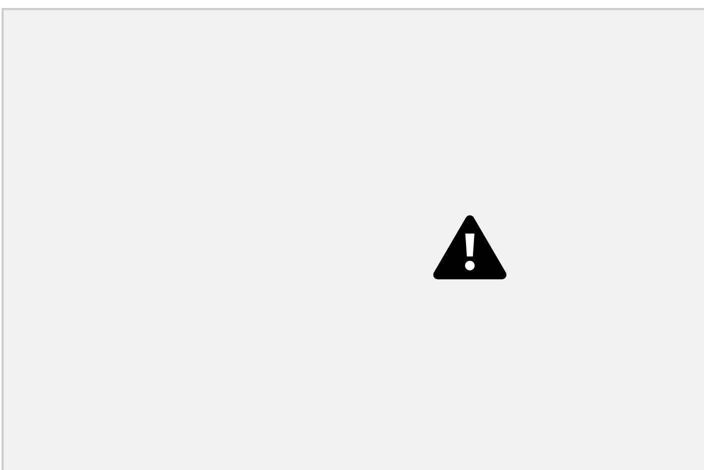
A liquid with a lower density will float on a liquid with a higher density if the liquids do not mix



Lower density liquids float on higher density liquids as long as the liquids do not mix

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Worked Example

Liquid A has a density of 0.76 g/cm^3 and liquid B has a density of 0.93 g/cm^3.

If the two liquids do not mix, which liquid will float on top of the other?

Step 1: List the known quantities

Liquid A = 0.76 g/cm^3

Liquid B = 0.93 g/cm^3

Step 2: Determine which liquid has the lowest density

The liquid with the lowest density will float on top of the liquid with the higher density

0.76 is less than 0.93

Therefore, liquid A has the lowest density

Step 3: State your answer

Liquid A will float on top of liquid B

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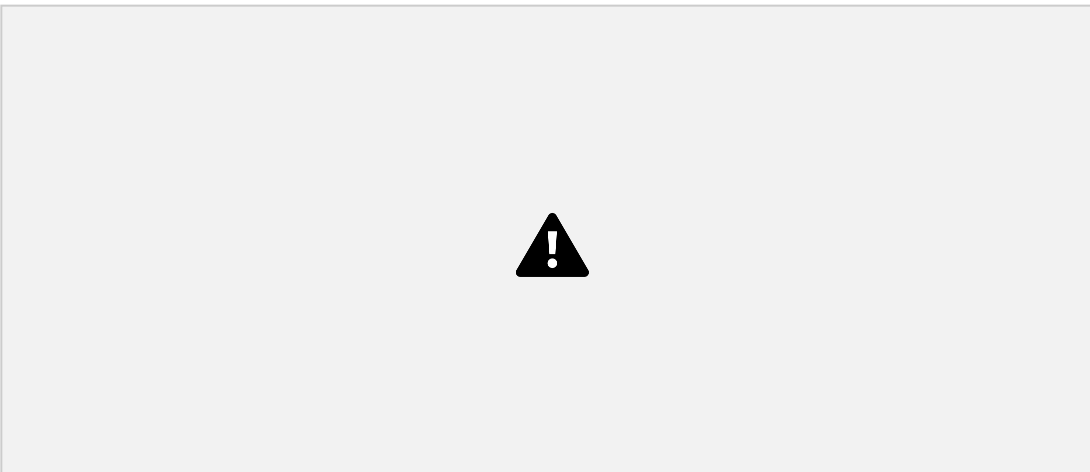
A force is defined as:

A push or a pull that acts on an object due to the interaction with another object

Forces can affect bodies in a variety of ways:

- Changes in **speed**: forces can cause bodies to speed up or slow down Changes in **direction**: forces can cause bodies to change their direction of travel
- Changes in **shape**: forces can cause bodies to **stretch**, **compress**, or **deform**



The effects of different forces on objects

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Resultant Forces on a Straight Line

A **resultant force** is a single force that describes all of the forces operating on a body

When many forces are applied to an object they can be combined (added) to produce one final force which describes the **combined action** of all of the forces This single resultant force determines:

The **direction** in which the object will move as a result of all of the forces The **magnitude** of the final force experienced by the object

The resultant force is sometimes called the **net force**

Forces can combine to produce

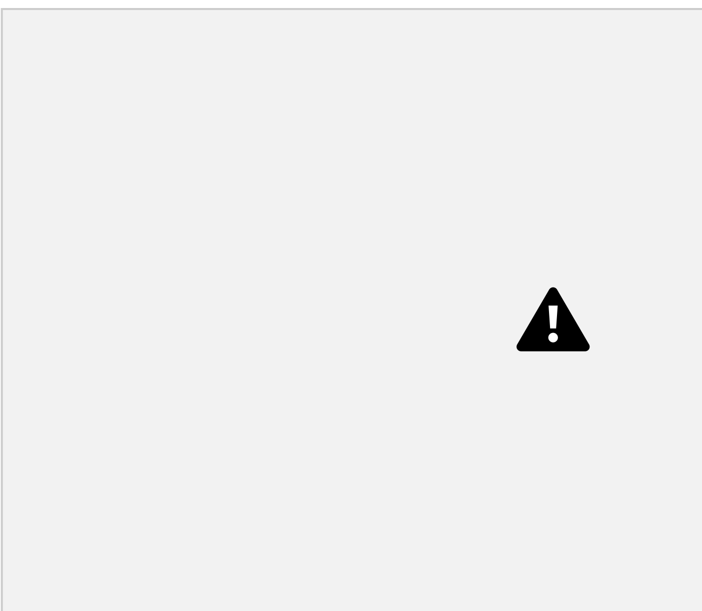
Balanced forces

Unbalanced forces

Balanced forces mean that the forces have combined in such a way that they cancel each other out and no resultant force acts on the body

For example, the **weight** of a book on a desk is balanced by the **normal** force of the desk

As a result, **no resultant force** is experienced by the book, the book and the table are **equal** and **balanced**



A book resting on a table is an example of balanced forces

Unbalanced forces mean that the forces have combined in such a way that they do not cancel out completely and there is a **resultant force** on the object For example, imagine two people playing a game of tug-of-war, working against each other on opposite sides of the rope

If person **A** pulls with 80 N to the left and person **B** pulls with 100 N to the right, these forces do not cancel each other out completely

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Since person **B** pulled with more force than person **A** the forces will be unbalanced and the rope will experience a resultant force of 20 N to the right

A tug-of-war is an example of when forces can become unbalanced

Resultant forces can be calculated by adding or subtracting all of the forces acting on the object

Forces working in opposite directions are **subtracted** from each other Forces working in the same direction are **added** together

If the forces acting in opposite directions are equal in size, then there will be no resultant force – the forces are said to be **balanced**

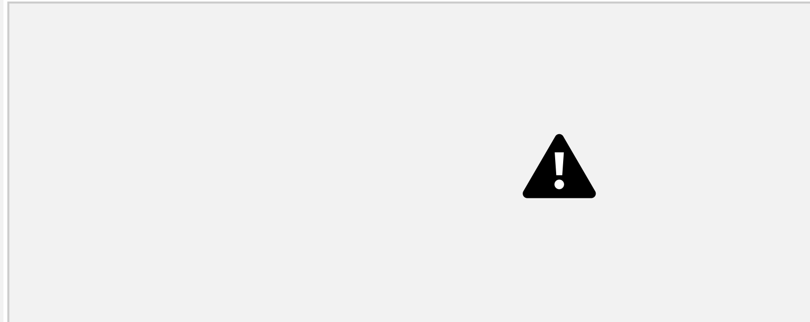

Diagram showing the resultant forces on three different objects

Imagine the forces on the boxes as two people pushing on either side In the first scenario, the two people are evenly matched - the box **doesn't move**

In the second scenario, the two people are pushing on the same side of the box, it moves to the right with their combined strength

In the third scenario, the two people are pushing against each other and are not evenly matched, so there is a **resultant force to the left**

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Worked Example

Calculate the magnitude and direction of the resultant force in the diagram below.

Step 1: Add up all of the forces directed to the right

4 N + 8 N = 12 N

Step 2: Subtract the forces on the right from the forces on the left 14 N – 12 N = 2 N

Step 3: Evaluate the direction of the resultant force

The force to the left is **greater** than the force to the right therefore the resultant force is directed to the left

Step 4: State the magnitude and direction of the resultant force The resultant force is 2 N to the left

Exam Tip

Remember to always provide units for your answer and to state whether the force is to the left, to the right, or maybe up or down

Always provide your final answer as a description of the **magnitude** and the **direction**, for example:

Resultant Force = **4 N to the right**

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Newton's First Law of Motion

Newton's **first law of motion** states:

Objects will remain at rest, or move with a constant velocity unless acted on by a resultant force

This means if the resultant force acting on an object is zero:

- The object will **remain stationary** if it was stationary before
- The object will continue to move at the **same velocity** if it was moving

When the resultant force is **not** zero

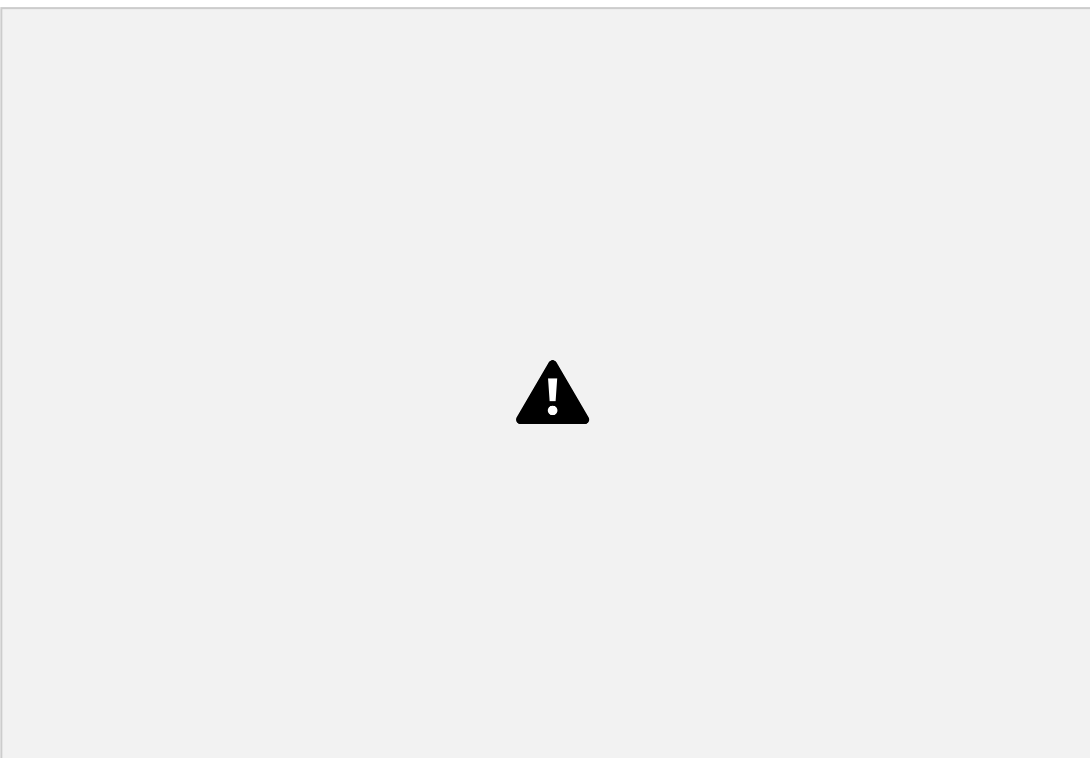
- The **speed** of the object can change
- The **direction** of the object can change

Applying Newton's First Law

Newton's first law is used to explain why things move with a **constant** (or **uniform**) **velocity**

- If the **forces** acting on an object are **balanced**, then the **resultant force** is **zero**
- The velocity (i.e. speed and direction) **can only change** if a **resultant force** acts on the object

A few examples with uniform velocity are shown below:



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Constant velocity can only be achieved when the forces on an object are balanced - in other words, when the resultant force is zero

Worked Example

Lima did some online research and found out that the Moon orbits the Earth at a constant speed of around 2000 mph.She says that this is not an example of Newton's first law of motion.Is Lima correct? Explain your answer.

Step 1: Recall Newton's first law of motion

Newton's first law of motion states that **objects will remain at rest, or move with a constant velocity, unless acted on by a resultant force**

Step 2: Determine if the object in the question is at rest, or if it is moving with a constant velocity

The Moon, in this case, is **not** at rest
It is moving at a **constant speed**
But it is **not** moving in a **constant direction** - it continually orbits the Earth Hence, it is not moving with a constant velocity, because velocity is a vector quantity

Step 3: State and explain whether Lima is correct

Lima is **correct**
The Moon moves with a **constant speed**, but always **changes direction** So it is **not** moving with a **constant velocity**, and is **not** an example of Newton's first law of motion

Worked Example

If there are no external forces acting on the car and it is moving at a constant velocity, what is the value of the frictional force, F?

Step 1: Recall Newton's first law of motion

Newton's first law of motion states that **objects will remain at rest, or move with a constant velocity unless acted on by a resultant force**

Step 2: Relate Newton's first law to the scenario

Since the car is moving at a constant velocity, there is no resultant force This means the driving and frictional forces are balanced

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force, F = driving force = **3 kN**

Step 3: State the value of the frictional force Frictional

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Newton's Second Law

Newton's **second law of motion** states:

> **The acceleration of an object is proportional to the resultant force acting on it and inversely proportional to the object's mass**

Newton's second law explains the following important principles:

- An object will **accelerate** (change its velocity) in response to a **resultant force**
- The **bigger** this resultant force, the **larger** the acceleration
- For a given force, the **greater** the object's mass, the **smaller** the acceleration experienced

The image below shows some examples of Newton's second law in action:

Objects like baseballs and lawnmowers accelerate when a resultant force is applied on them. The size of the acceleration is proportional to the size of the resultant force

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Calculations Using Newton's Second Law

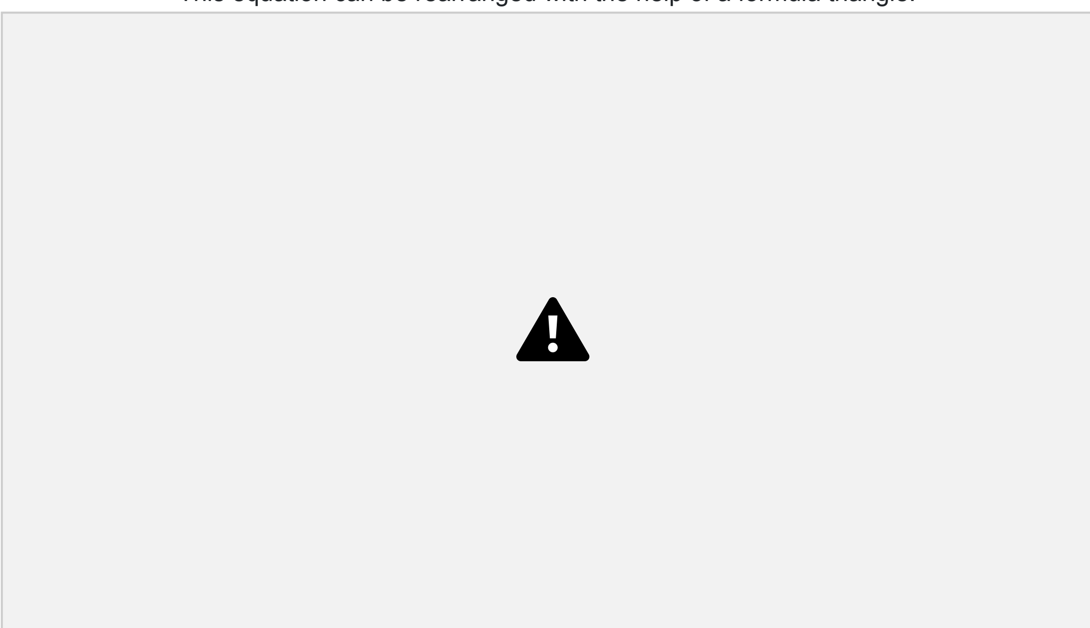
Newton's second law can be expressed as an equation:

F = ma

Where:
F = resultant force on the object in Newtons (N)
m = mass of the object in kilograms (kg)

a = acceleration of the object in metres per second squared (m/s^2) The **force** and the **acceleration** act in the **same direction**

This equation can be rearranged with the help of a formula triangle:



Force, mass, acceleration formula triangle

Worked Example

A car salesman says that his best car has a mass of 900 kg and can accelerate from 0 to 27 m/s in 3 seconds.

Calculate:

a) The acceleration of the car in the first 3 seconds.

b) The force required to produce this acceleration.

Part (a)

Step 1: List the known quantities

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Initial velocity = 0 m/s
Final velocity = 27 m/s
Time, t = 3 s

Step 2: Calculate the change in velocity

change in velocity = Δv = final velocity − initial velocity

Δv = 27 − 0 = 27 m/s

Step 3: State the equation for acceleration

Step 4: Calculate the acceleration

a = 27 ÷ 3 = **9 m/s**

Part (b)

Step 1: List the known quantities

Mass of the car, m = 900 kg

Acceleration, a = 9 m/s^2

Step 2: Identify which law of motion to apply

The question involves quantities of **force**, **mass** and **acceleration**, so Newton's second law is required:

F = ma

Step 3: Calculate the force required to accelerate the car

F = 900 × 9 = **8100 N**

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Worked Example

Three shopping trolleys, **A**, **B** and **C**, are being pushed using the same force. This force causes each trolley to accelerate.

Which trolley will have the smallest acceleration? Explain your answer.

Step 1: Identify which law of motion to apply

The question involves quantities of **force** and **acceleration**, and the image shows trolleys of different **masses**, so **Newton's second law** is required:

$$F = ma$$

Step 2: Re-arrange the equation to make acceleration the subject

Step 3: Explain the inverse proportionality between acceleration and mass

Acceleration is **inversely proportional** to mass

This means for the **same amount of force**, a **large mass** will experience a **small acceleration**

Therefore, trolley **C** will have the smallest acceleration because it has the largest mass

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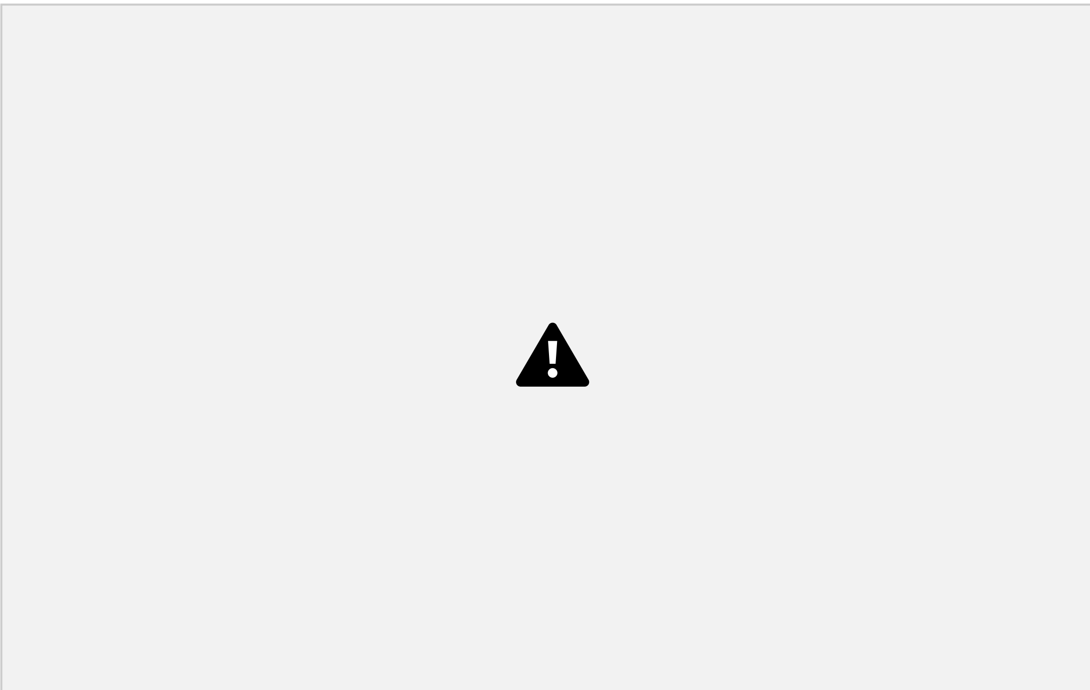
Investigating Springs

When forces are applied to materials, the size and shape of the material can change

The method below describes a typical procedure for carrying out an investigation into the properties of a material



An experiment to measure the extension of a spring

Set up the apparatus as shown in the diagram

A single mass (0.1 kg, 100g) is attached to the spring, with a pointer attached to the bottom, and the position of the spring is measured against the ruler

The mass (in kg) and position (in cm) are recorded in a table

A further mass is added and the new position measured

The above process continues until a total of 7 masses have been added

The masses are then removed and the entire process repeated again, until it has been carried out a total of three times, and averages can then be taken

Once measurements have been taken:

- The force on the spring can be found by multiplying the mass on the spring (in kg) by 9.81 N/kg (the gravitational field strength)
- The extension of the spring can be found by subtracting the original position of the spring from each of the subsequent positions

Finally, a graph of extension (on the y-axis) against force (on the x-axis) should be plotted

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A graph of force against extension for a metal spring

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